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Corona

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(54) **TWO-PIECE MOISTURE RESISTANT LOCKING CONNECTOR WITH MULTI PIN CONNECTION**

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(73) Assignee: **DSM&T COMPANY INC.**, Fontana, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 313 days.

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Related U.S. Application Data

(63) Continuation-in-part of application No. 18/062,971, filed on Dec. 7, 2022.

(51) **Int. Cl.**
H01R 13/52 (2006.01)
H01R 13/627 (2006.01)

(52) **U.S. Cl.**
CPC ***H01R 13/5202*** (2013.01); ***H01R 13/5219*** (2013.01); ***H01R 13/6272*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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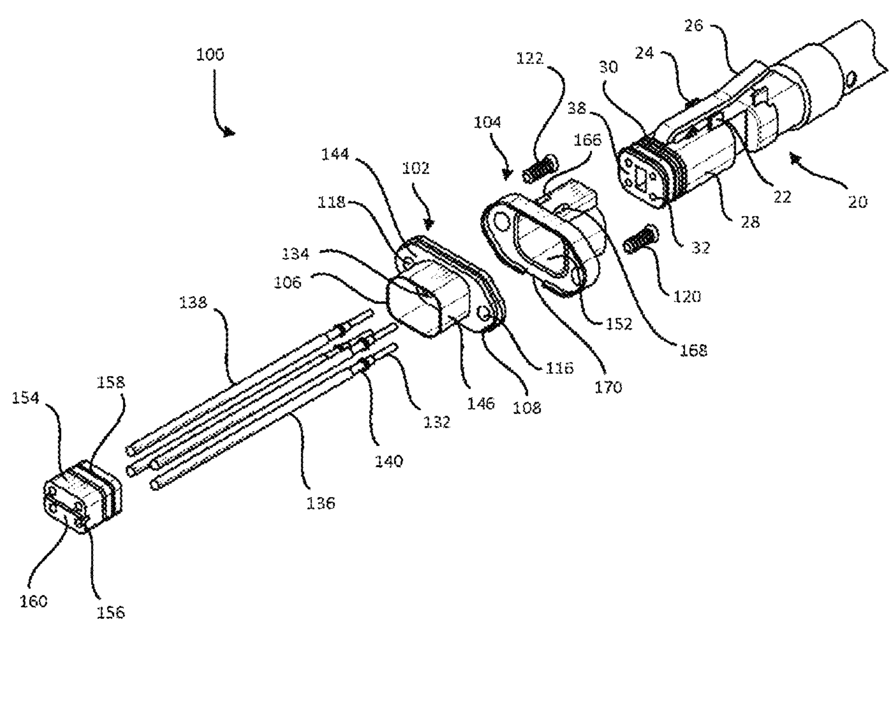
(57) **ABSTRACT**

A waterproof seal for electrical assemblies wherein at least two pins extend through the electrical conductor such that the at least two pins extending from a distal end of the electrical conductor and are adapted to interact with openings on a connecter that is detachably connectable with the distal end of the electrical connector, and the at least two pins extend from a proximal end of the electrical connector such that the at least two pins extending from the proximal end are adapted to be interact with openings in a socket formed in the surface of housing where the electrical connector forms a water tight barrier preventing water from entering the housing through the electrical connector or around the electrical connector.

32 Claims, 26 Drawing Sheets



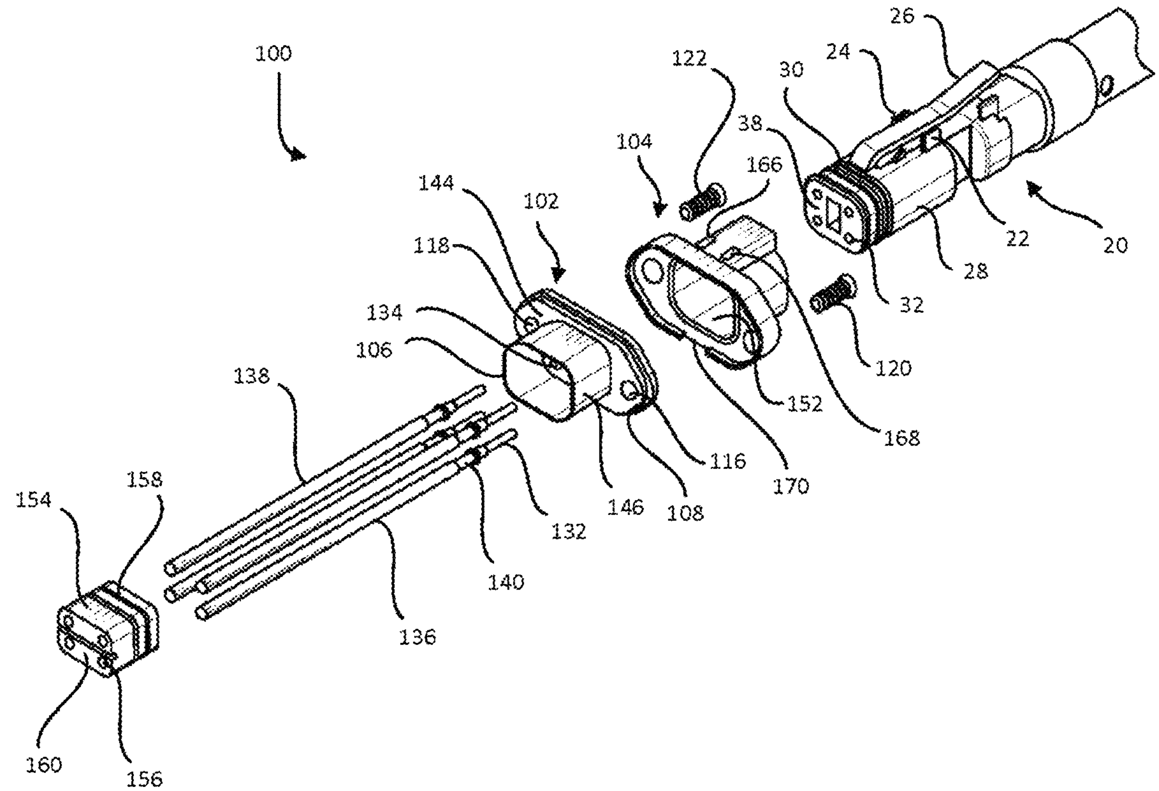

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FIG. 1

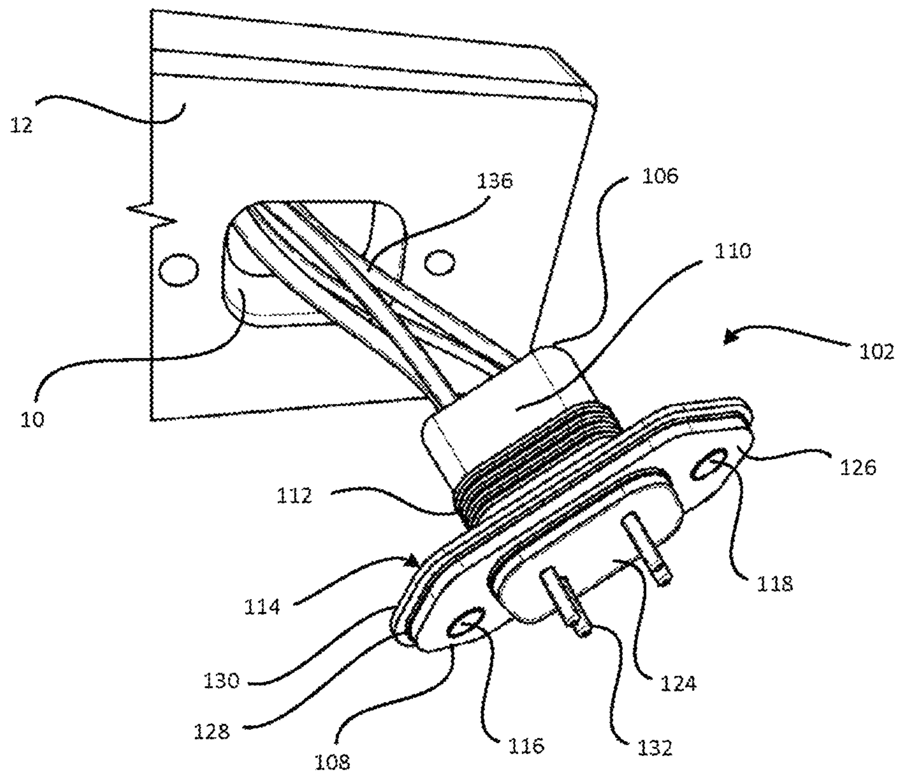
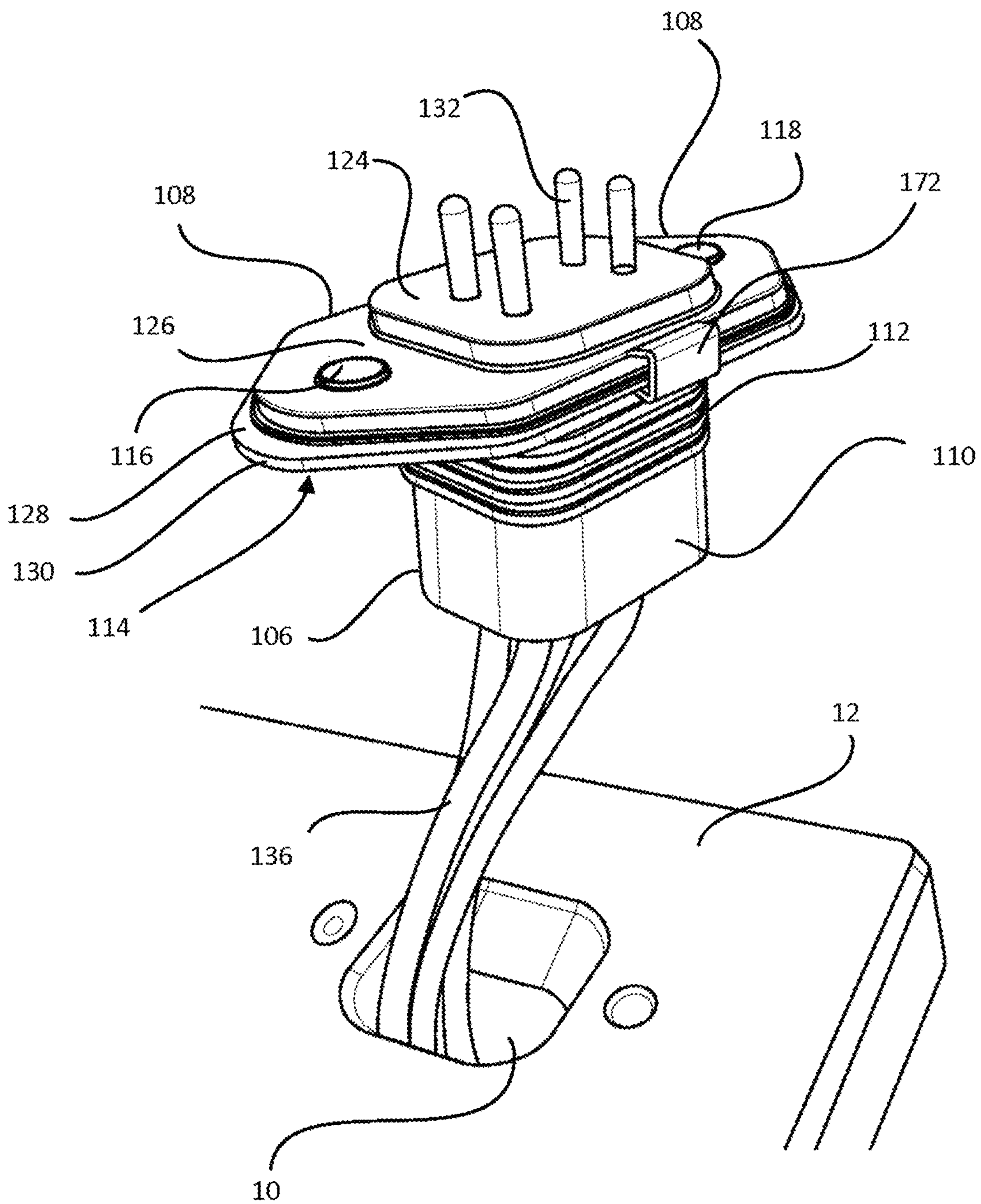


FIG. 2



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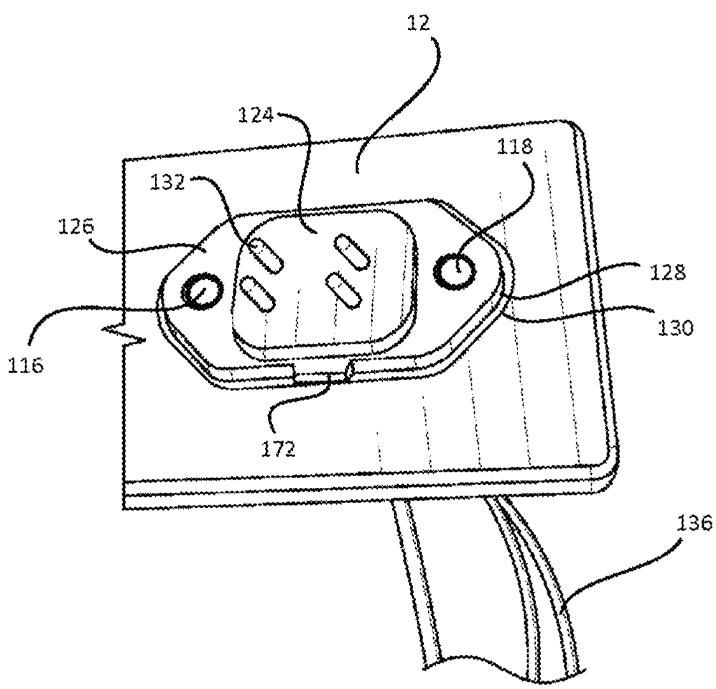

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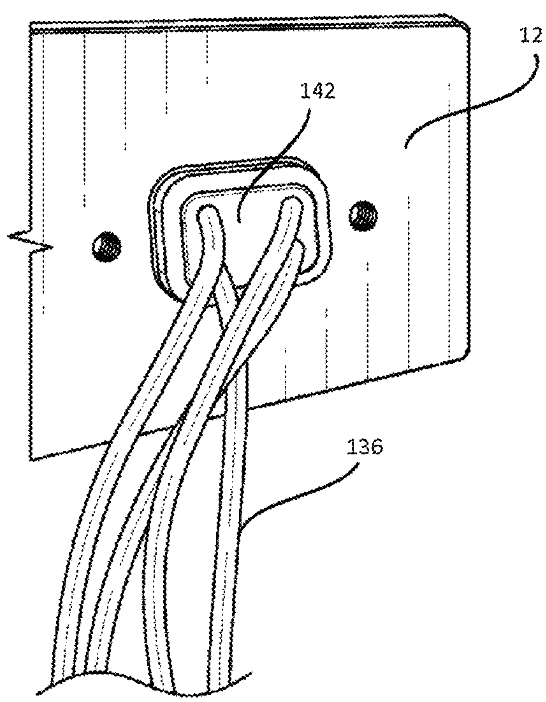

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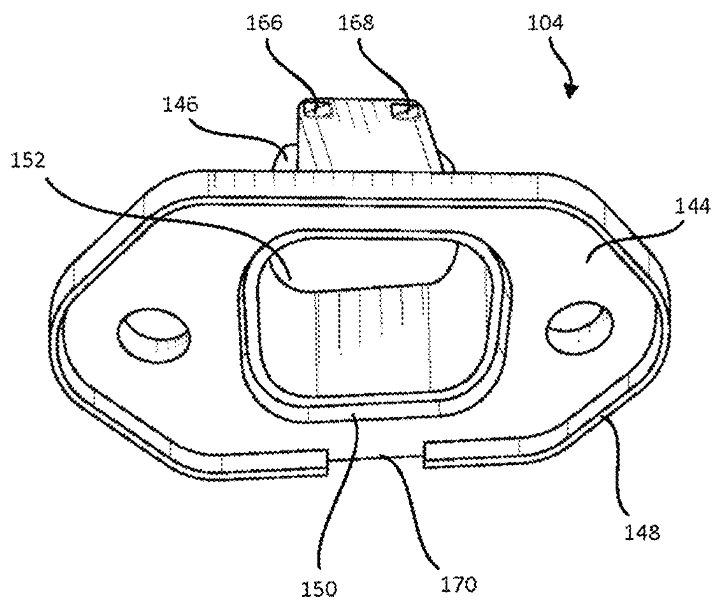

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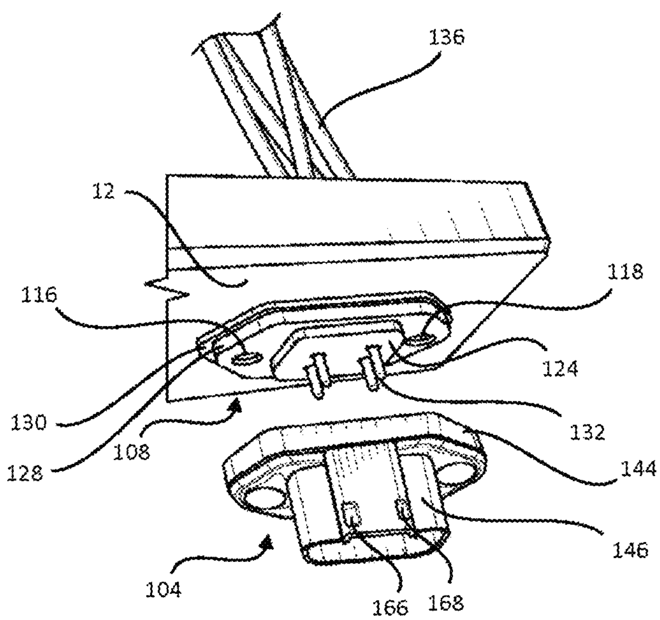

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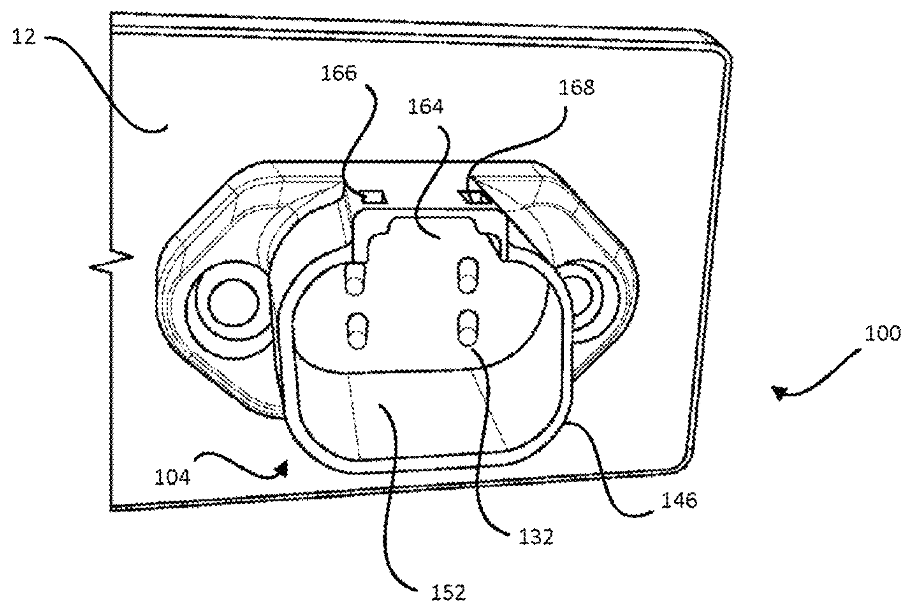

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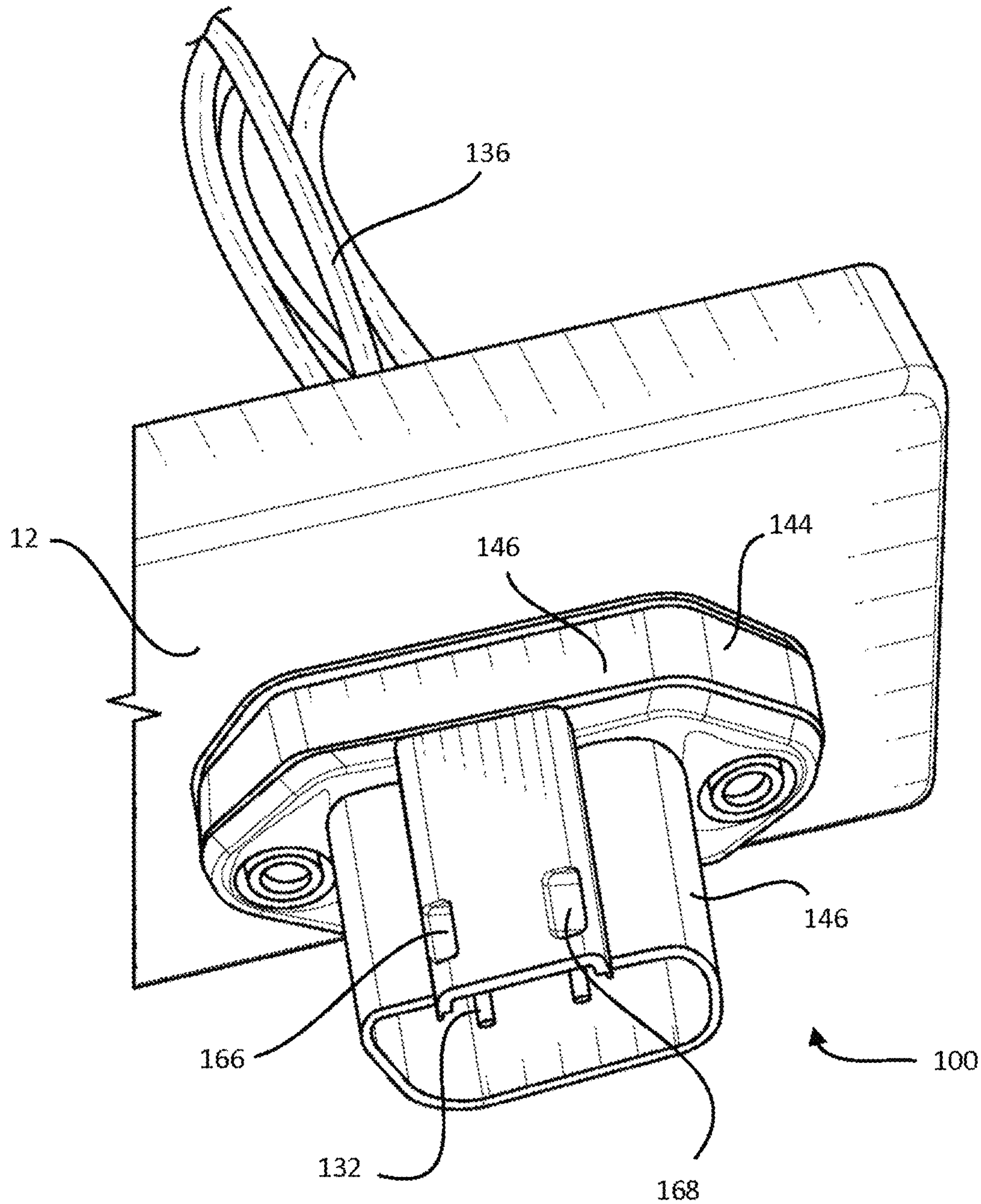

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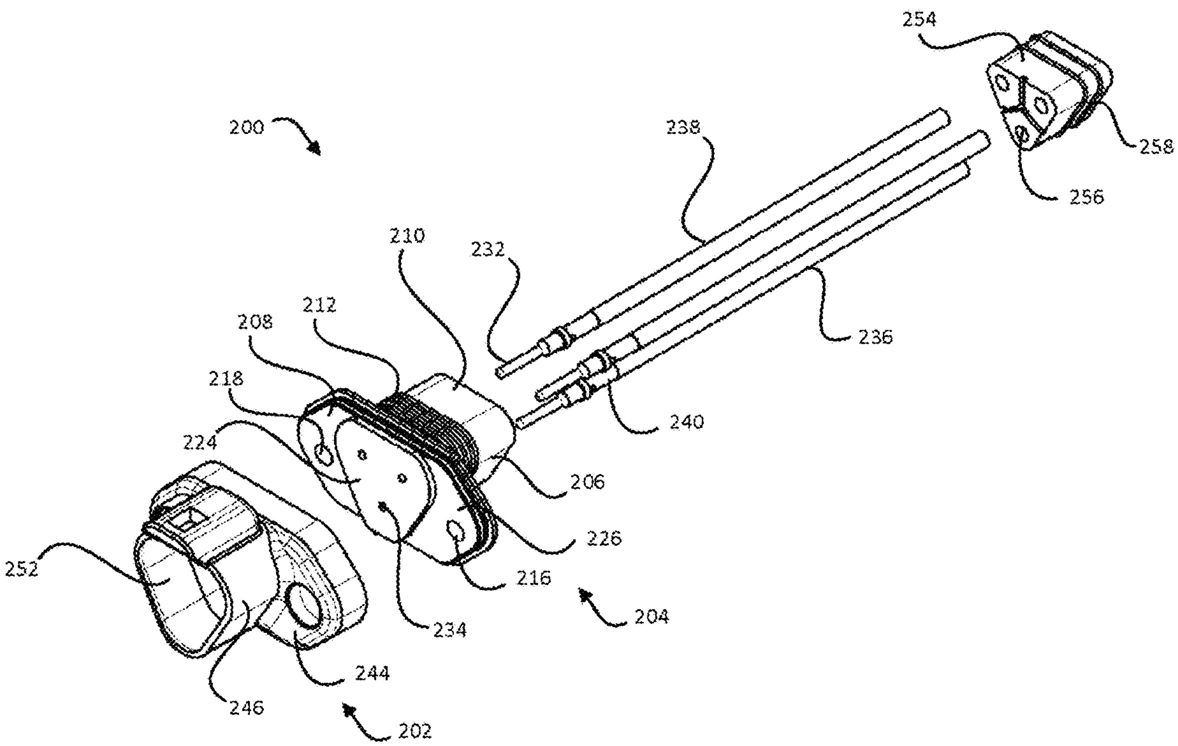
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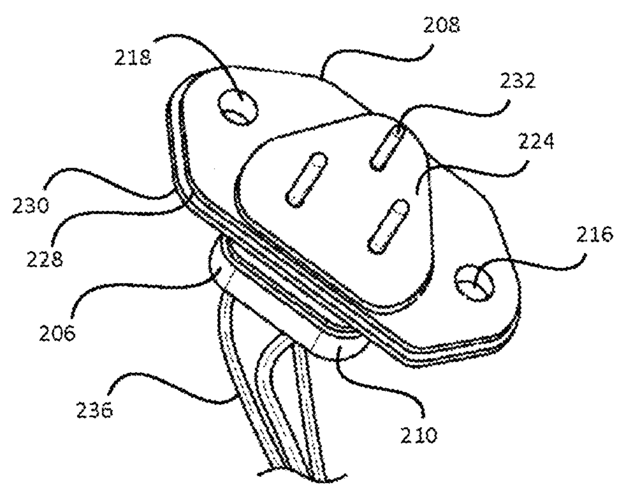
FIG. 10



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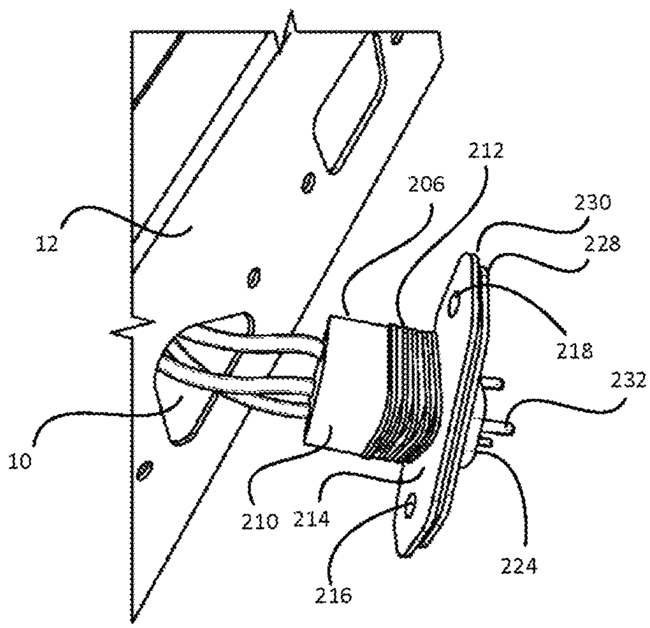

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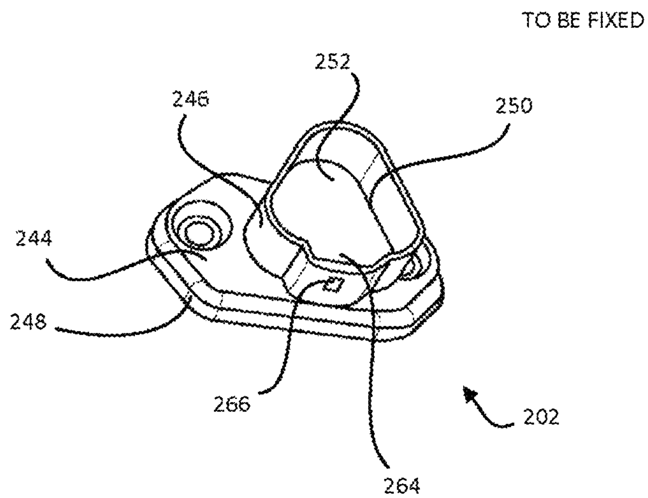

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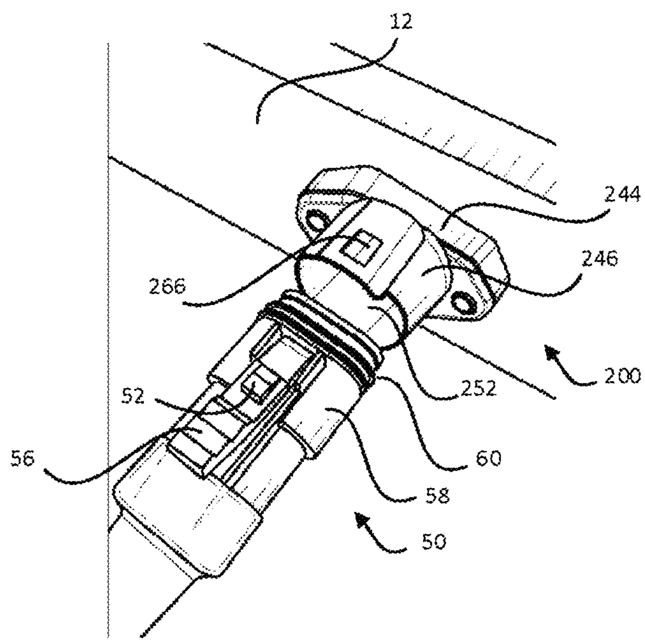

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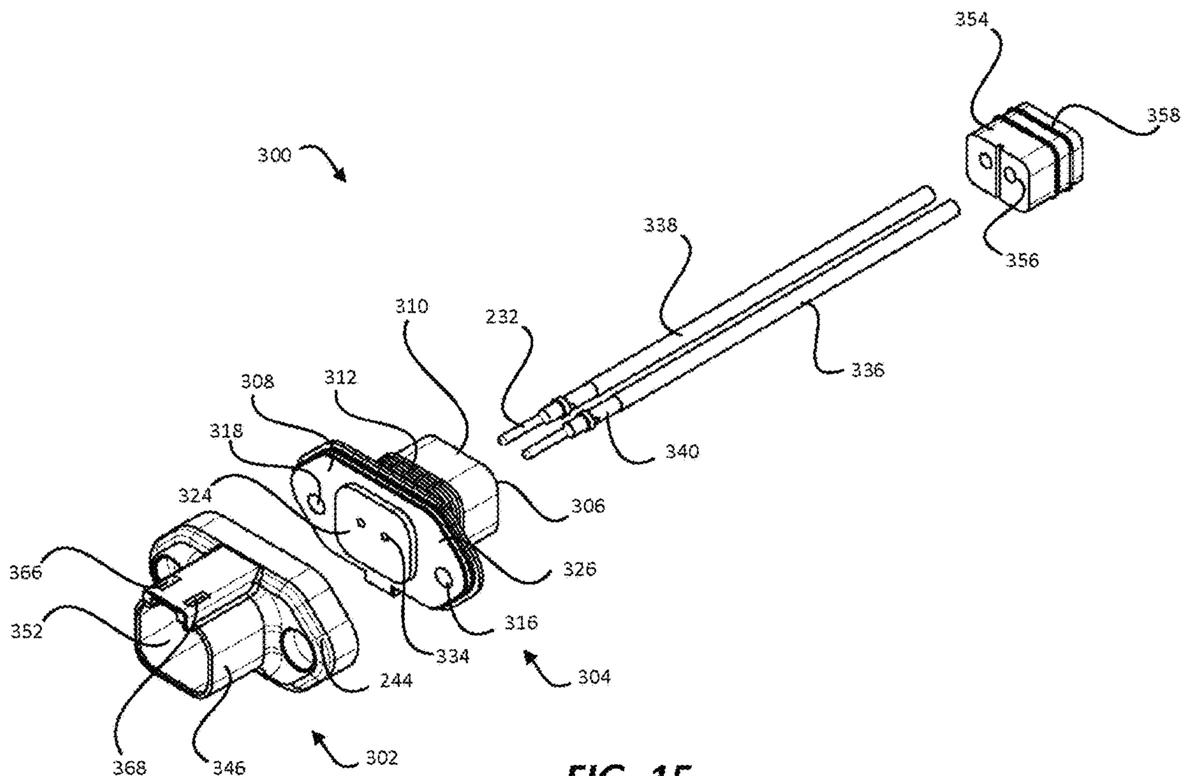
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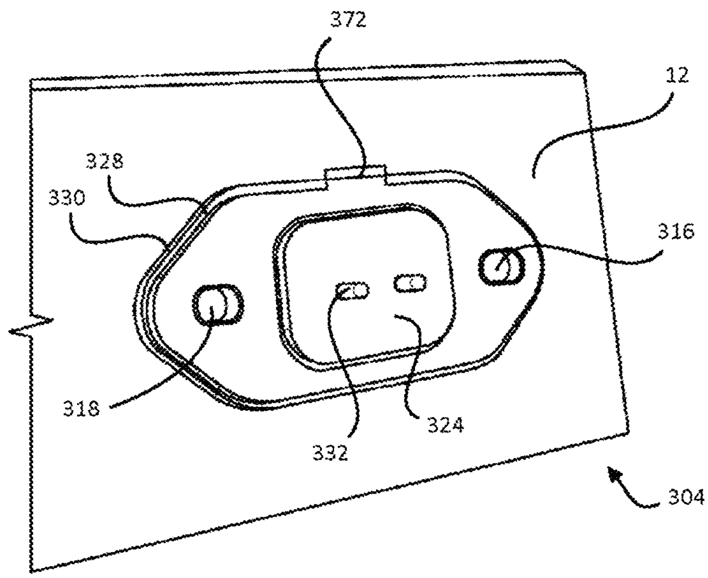
FIG. 15

TO BE FIXED



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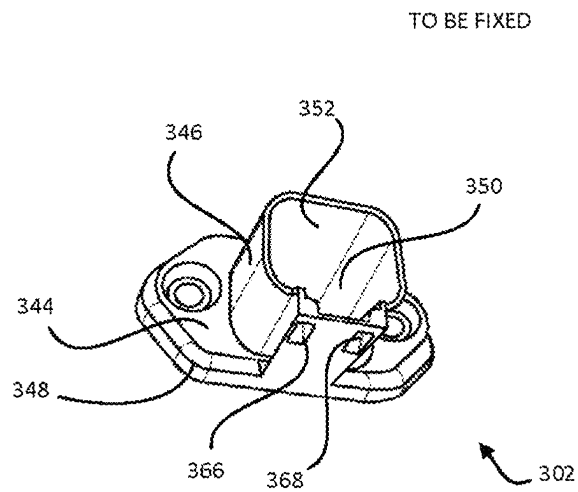

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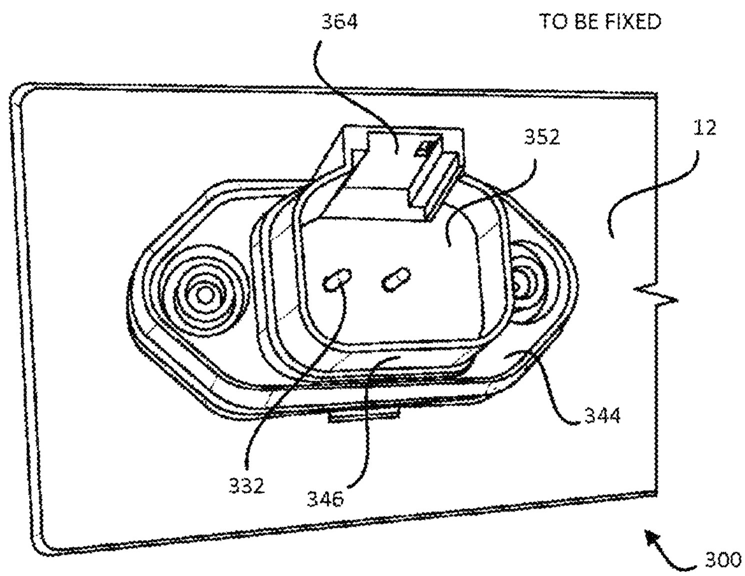

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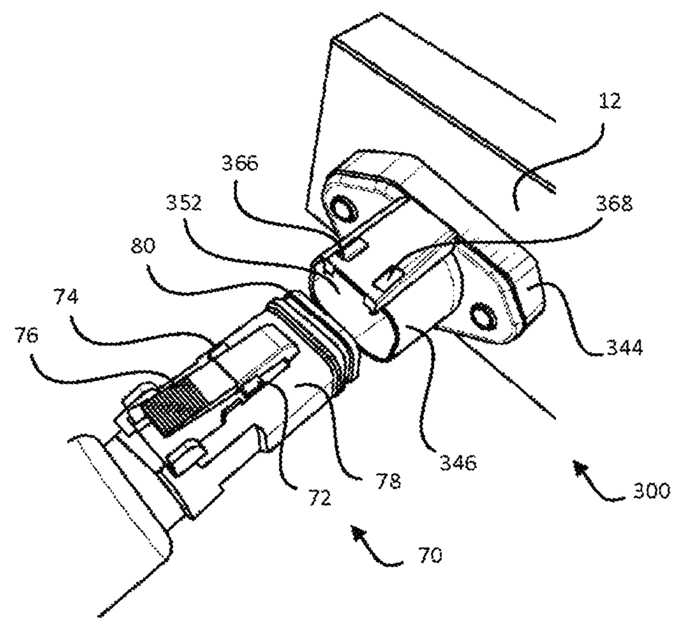

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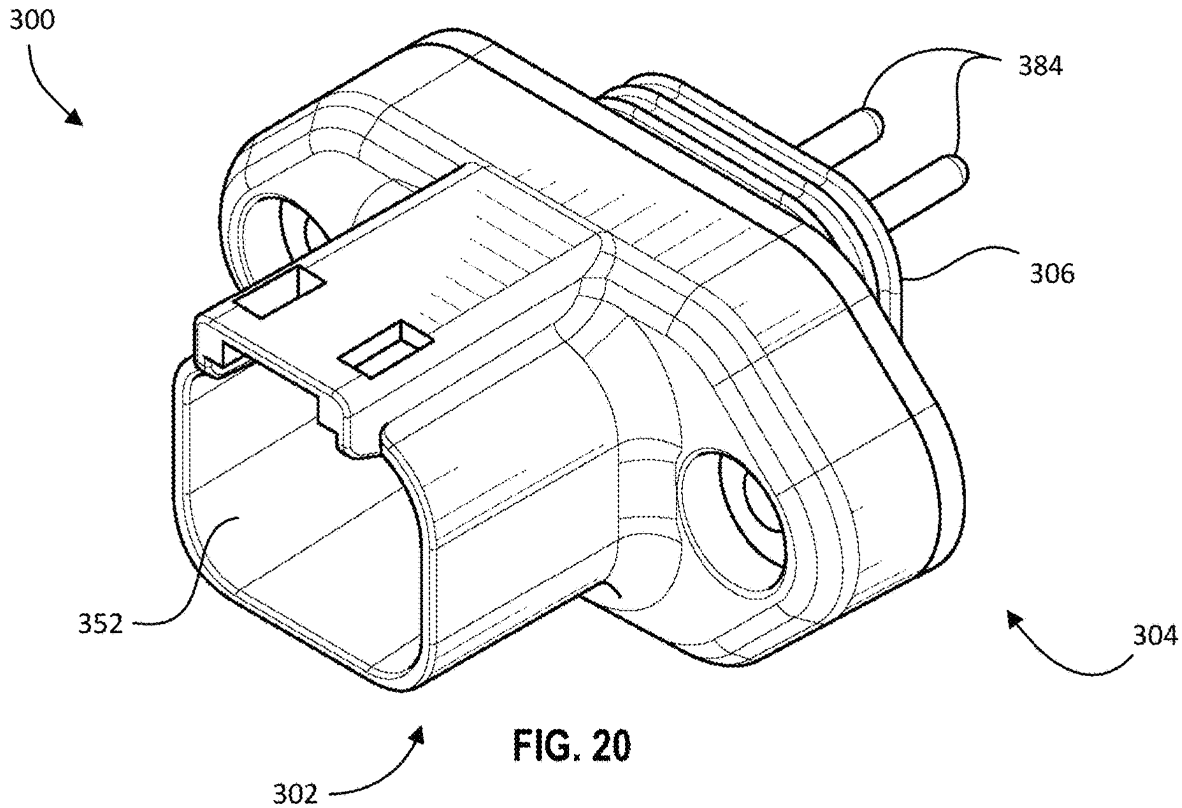

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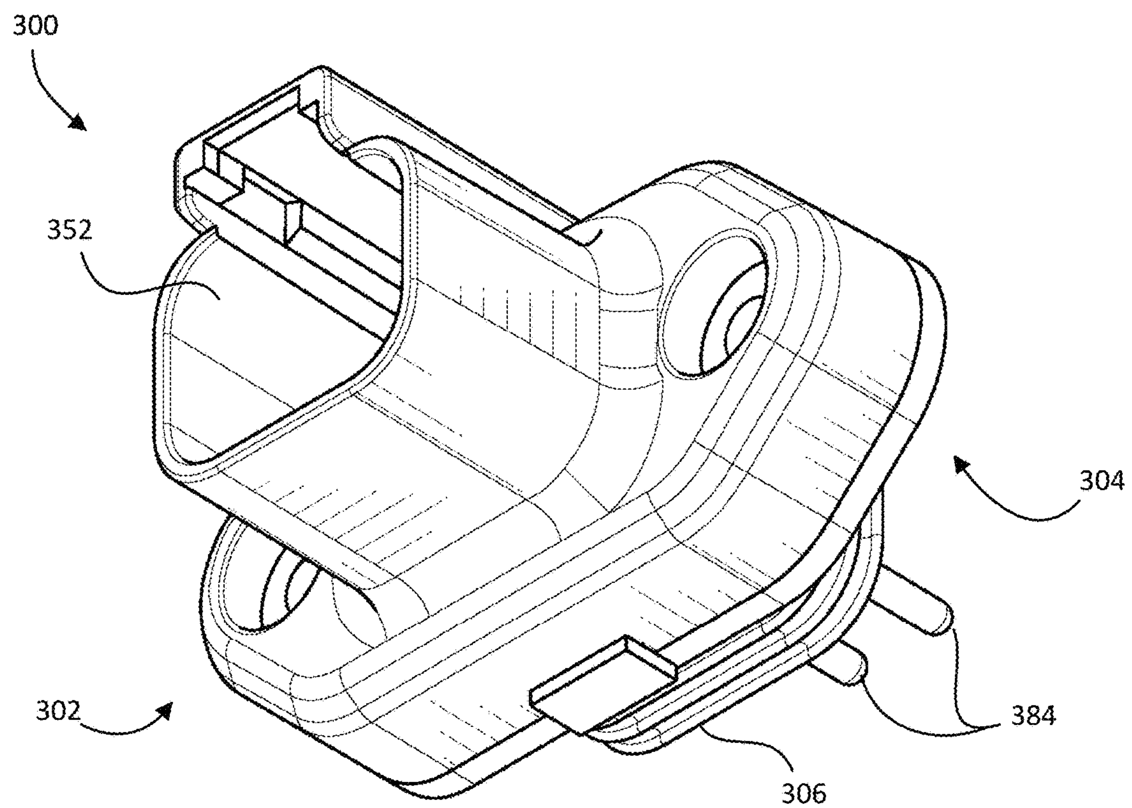

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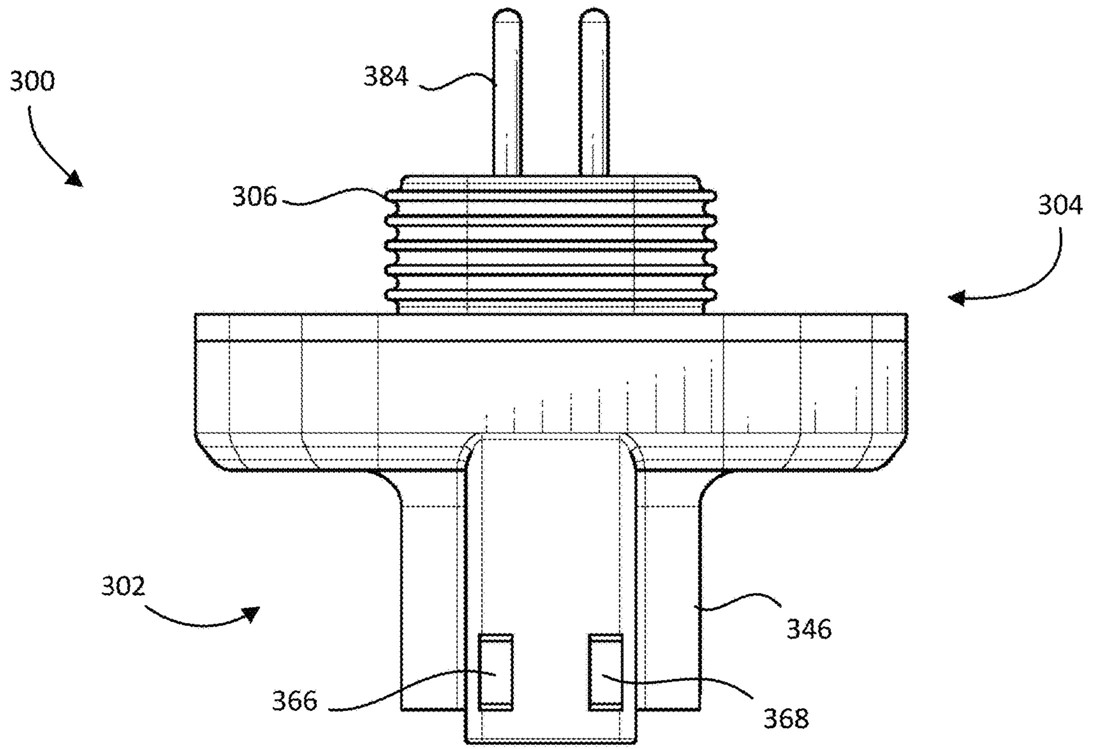
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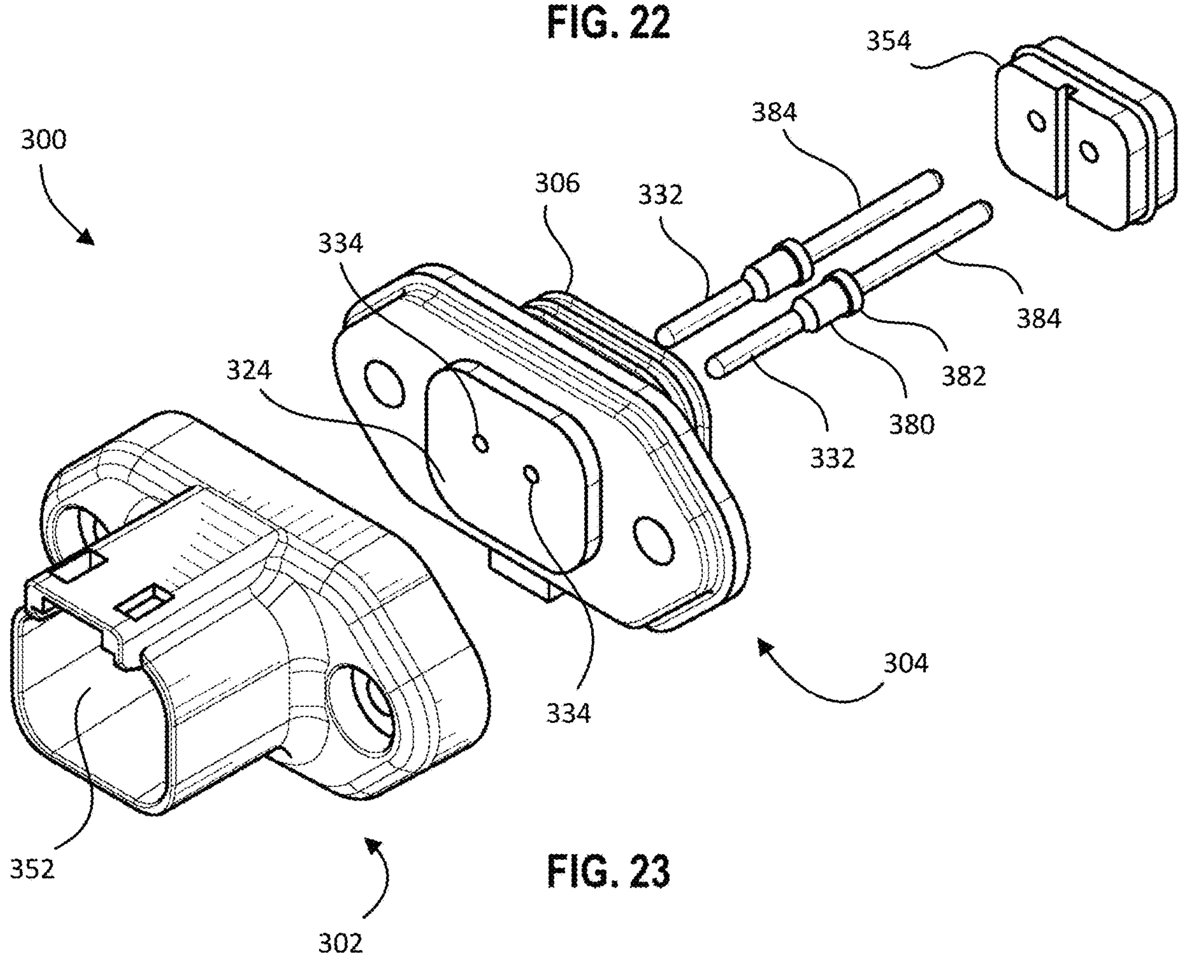
FIG. 22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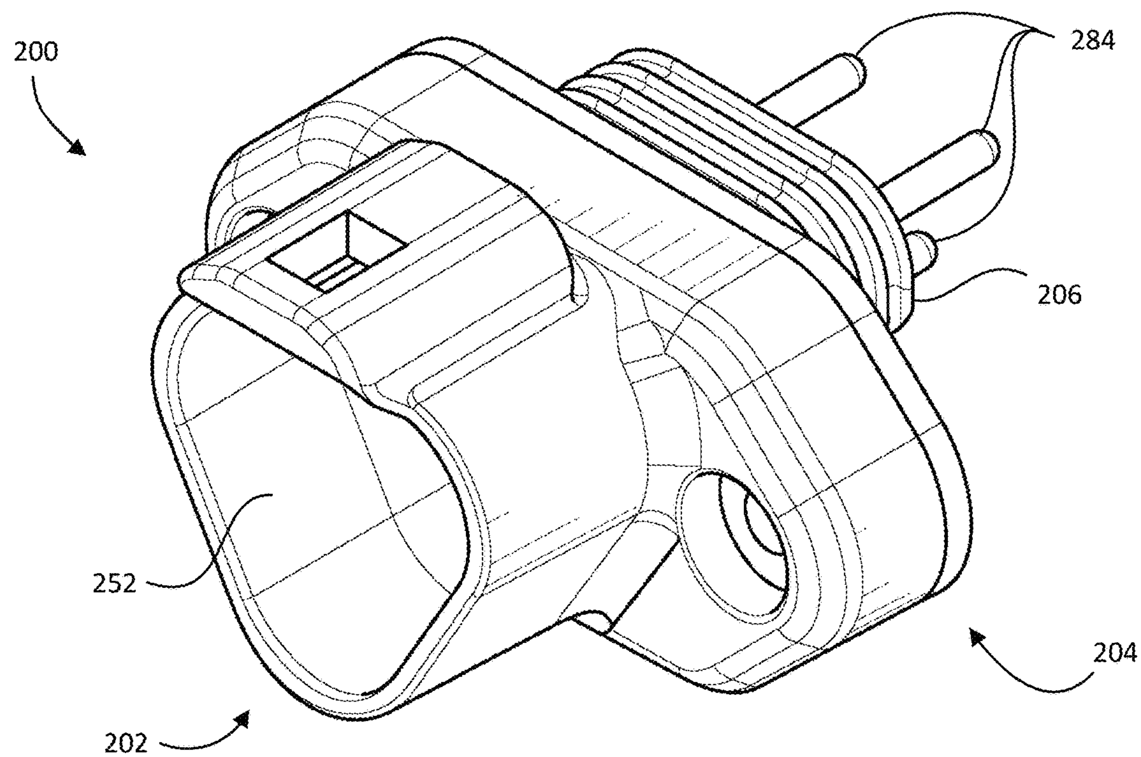

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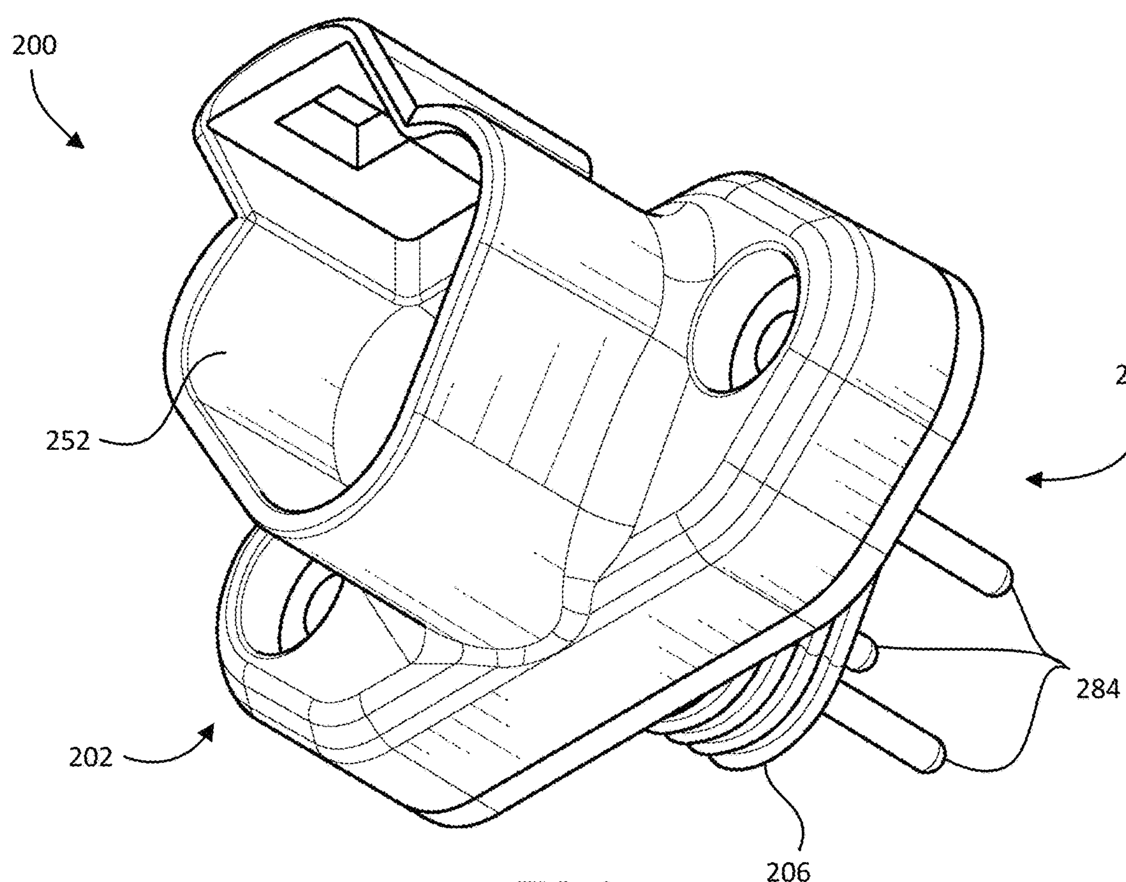

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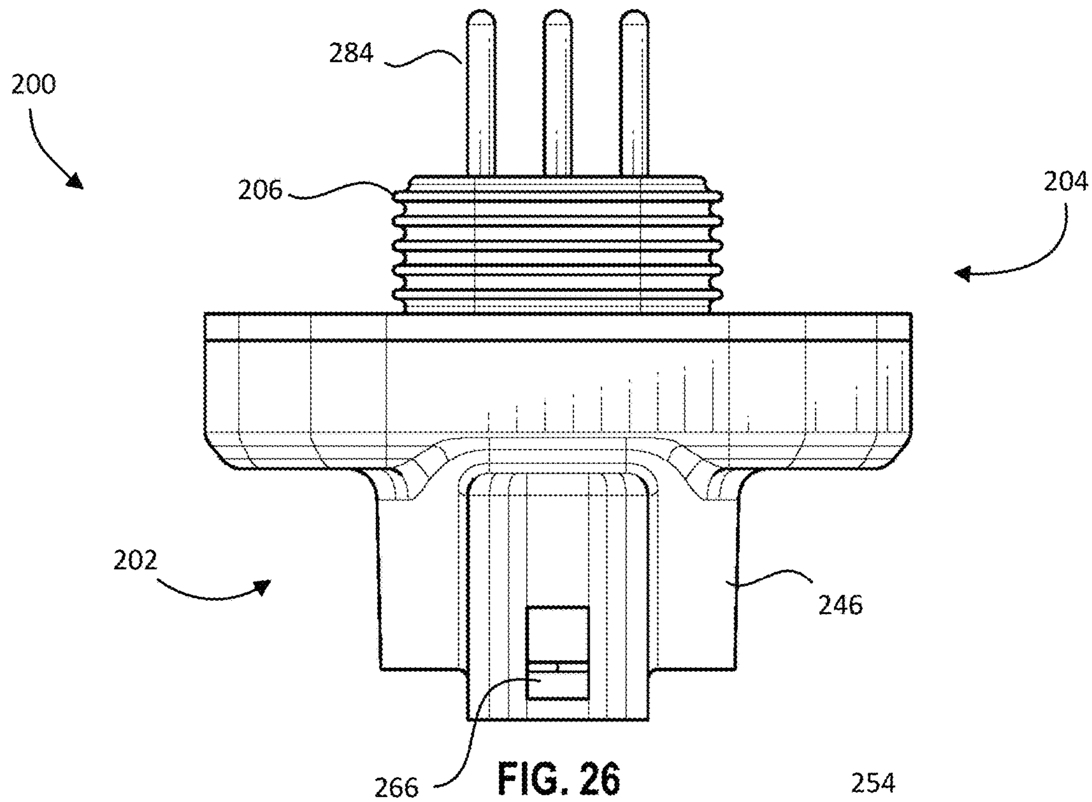
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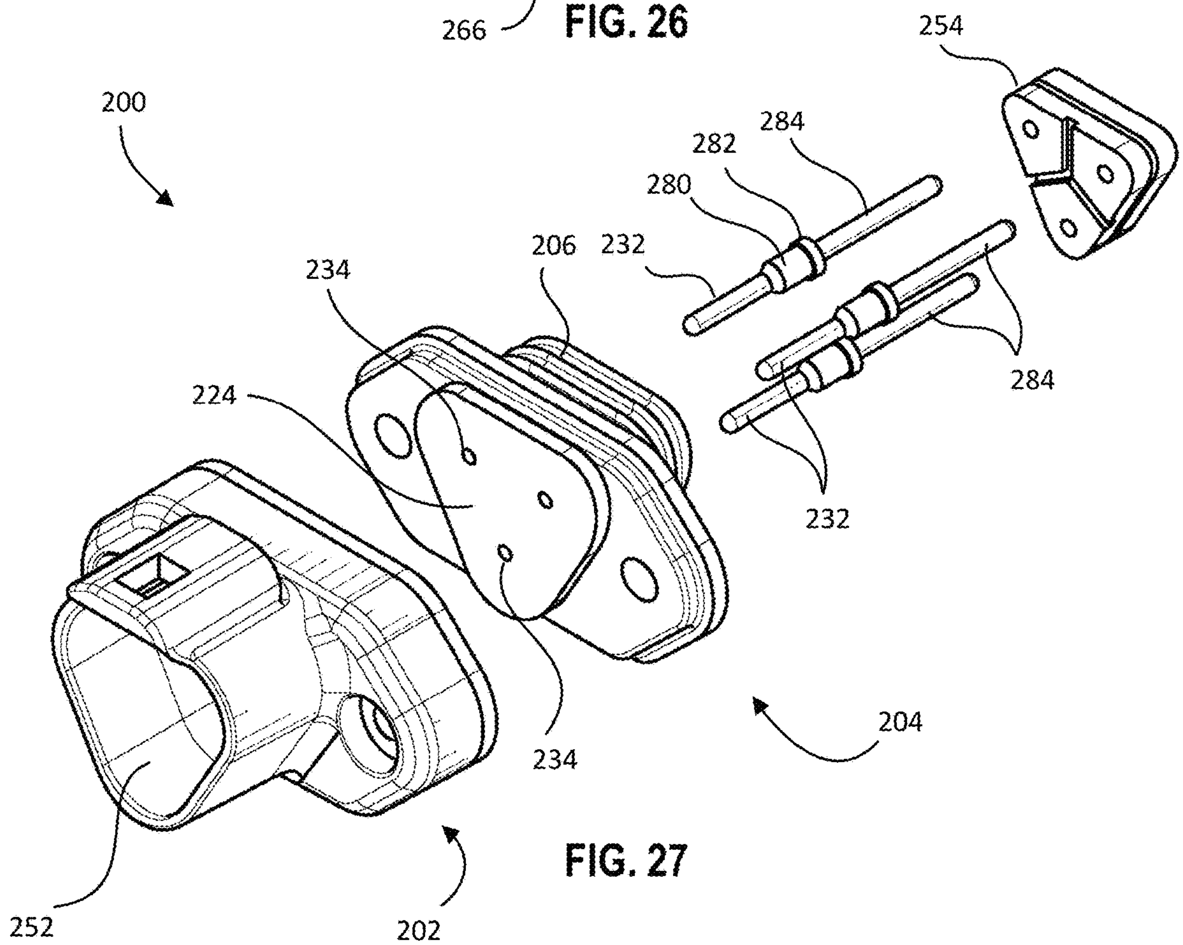
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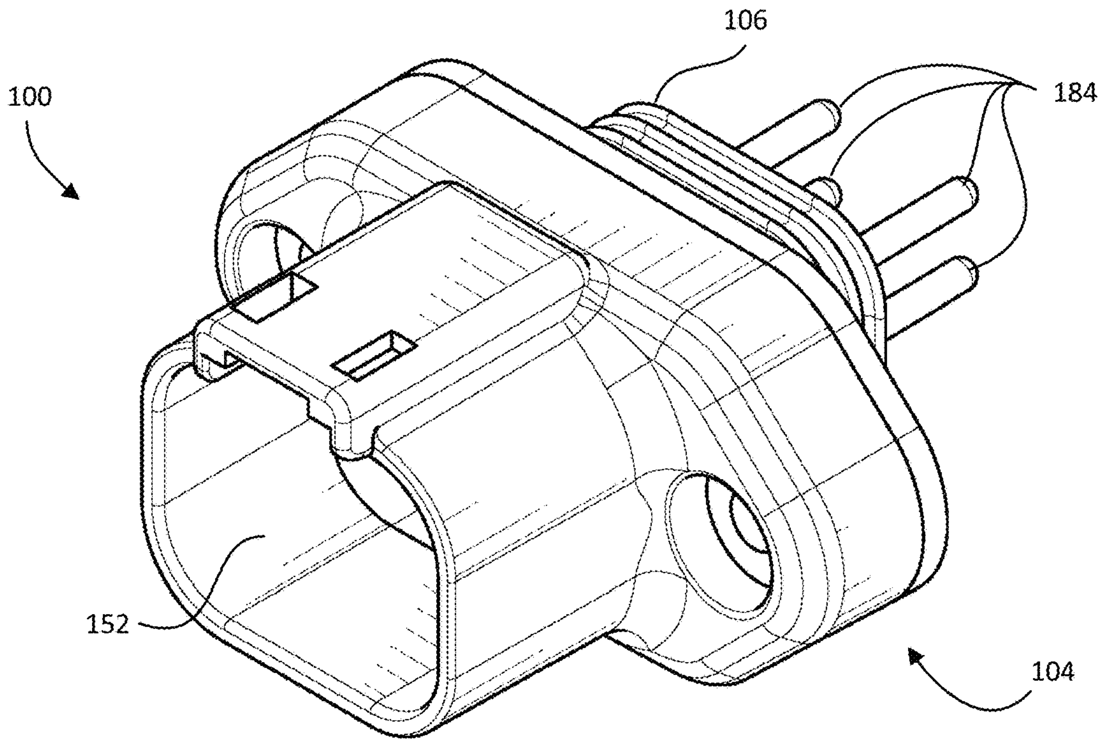

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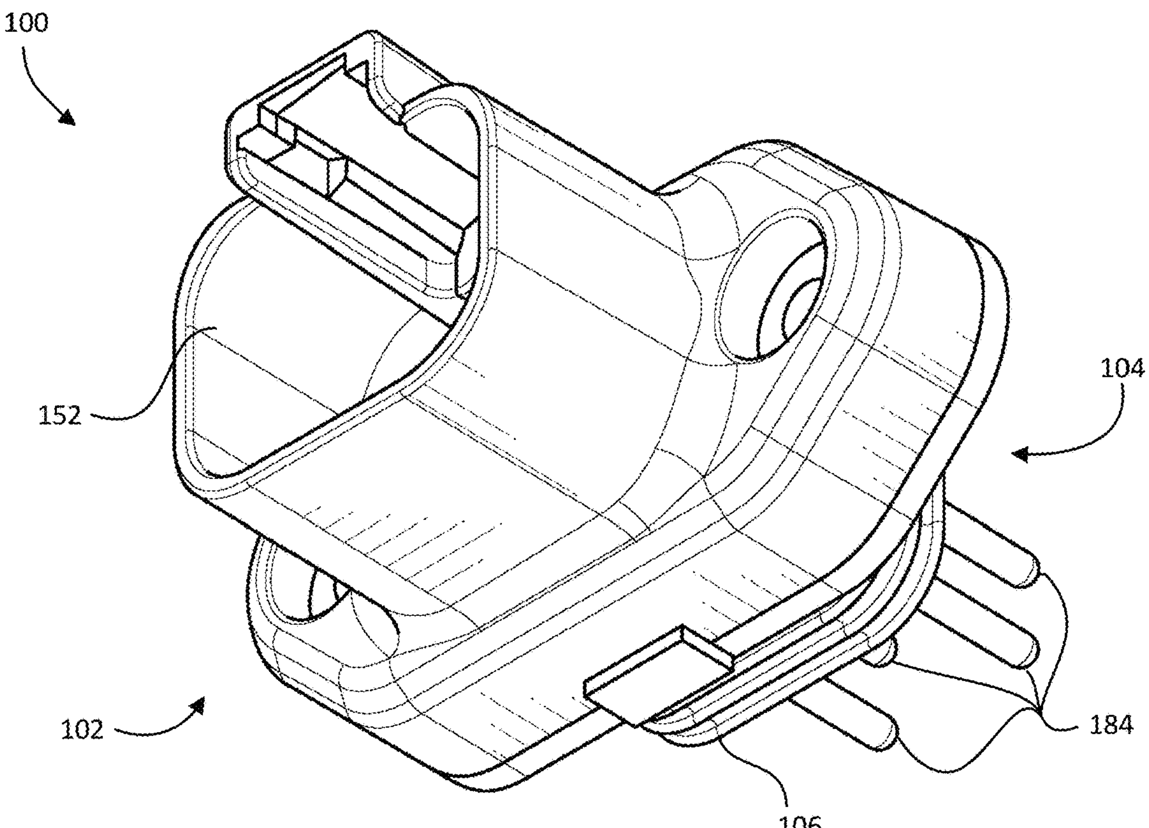

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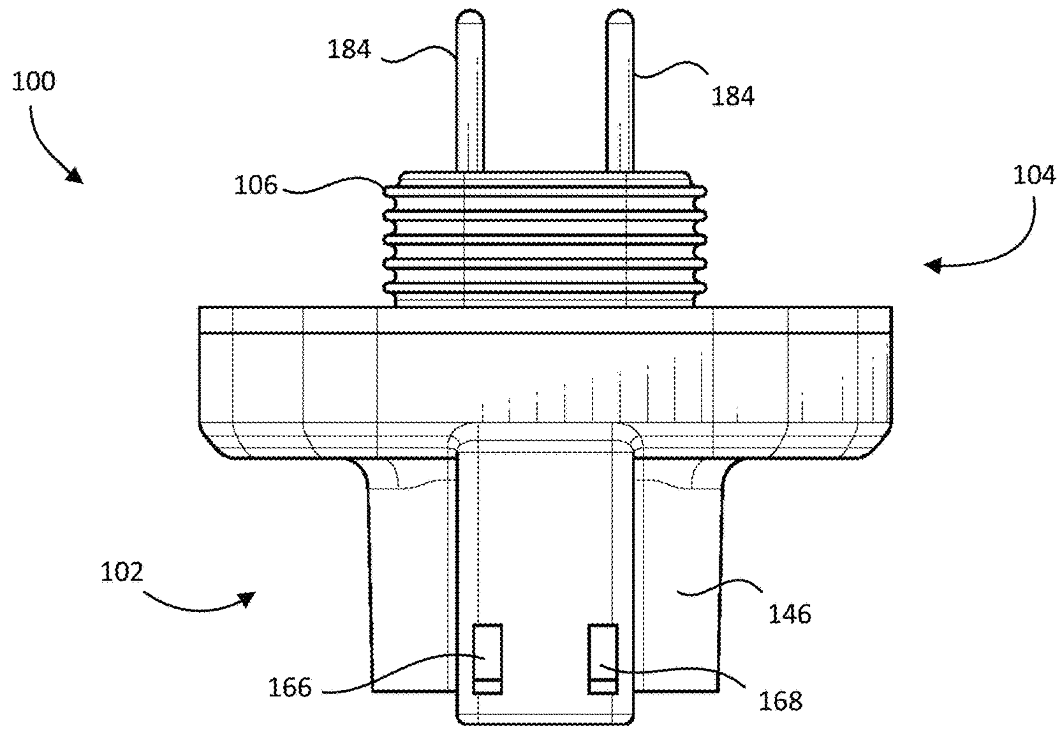

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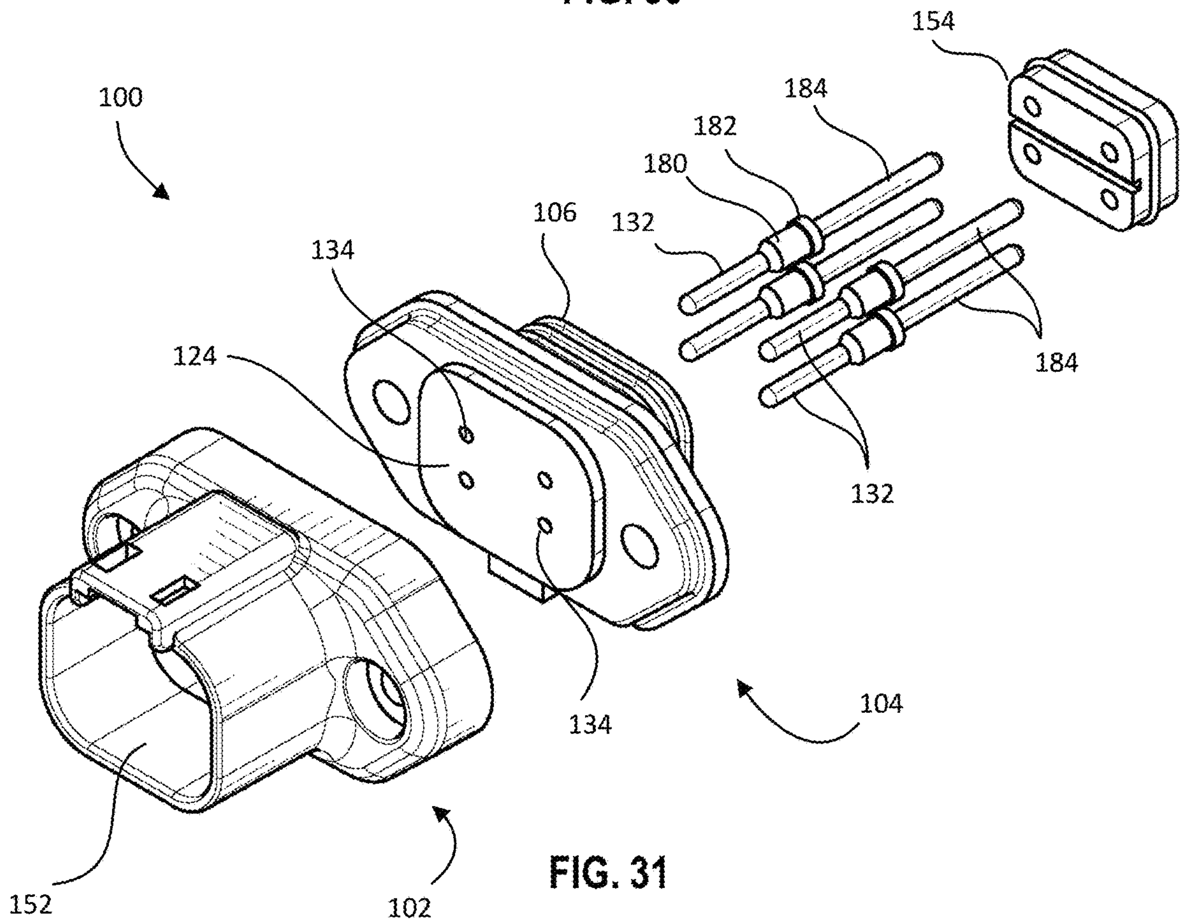

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FIG. 31

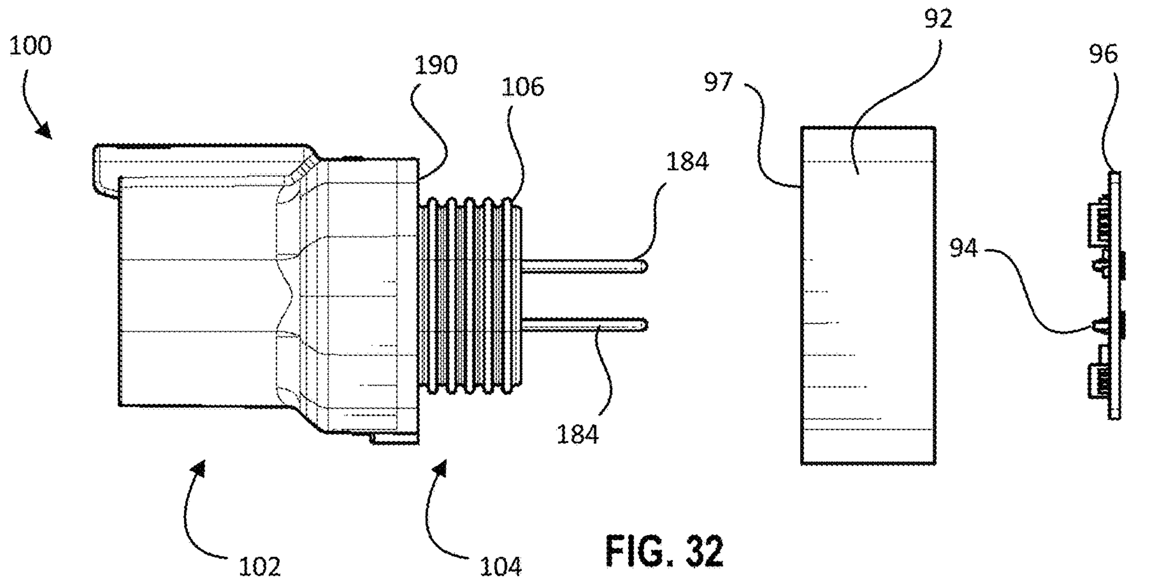
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FIG. 32

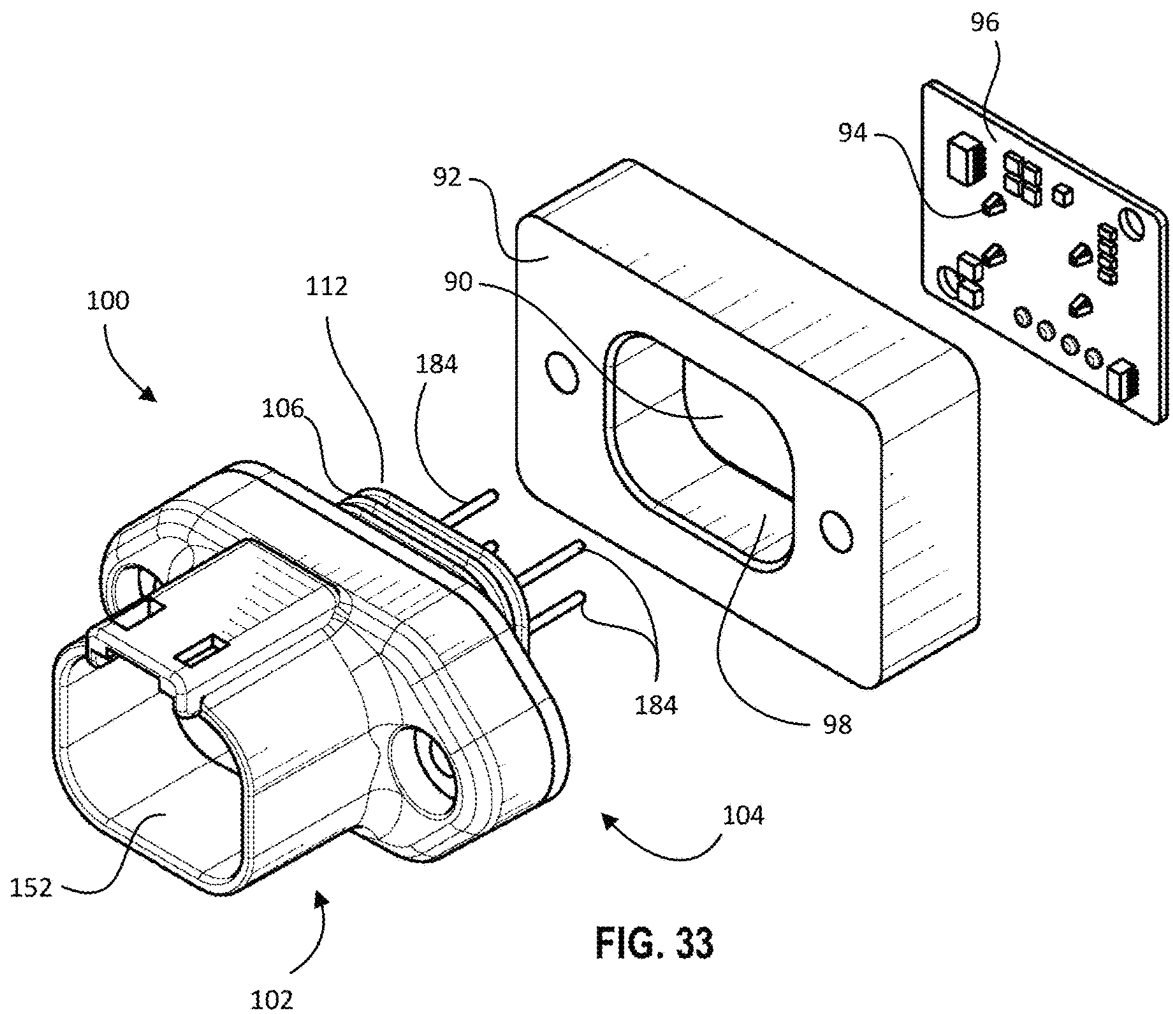
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FIG. 33

TWO-PIECE MOISTURE RESISTANT LOCKING CONNECTOR WITH MULTI PIN CONNECTION

FIELD OF THE INVENTION

The present invention relates to a waterproof electrical connector and more specifically, to a two-piece electrical connector that is designed to mate at one end with a Deutsch Connector and where the electrical connector is designed such that a second end is designed to mate with a receptacle and includes a plurality of seals including a detachable connector body that sits over top of the connector base and forms a cavity in which the Deutsch Connector is received.

BACKGROUND OF THE INVENTION

One challenge faced in the electrical lighting industry is the prevention of water intrusion into lighting fixtures and electronics. This is especially an issue for light fixtures designed to be in wet locations such as exterior lighting, areas with standing water, or automotive and marine applications. Lighting manufacturers go to great lengths to design weather tight enclosures to prevent water intrusion and various methods have been tried with limited success.

Many problems can arise when water gets into a light fixture or into electronics. One major problem is corrosion that can shorten the life of the light fixture or electronics. Other issues relate to safety, such as short-circuits and ground-faults, that can occur as the light fixture or electronics degrade. Still other issues include reduced performance for the equipment fed by the electrical conductors extending from a leaky electrical connector.

To deal with these issues, lighting manufacturers have sought to provide better moisture seals to completely seal off the interior space of light fixtures and electronics from the outside. While manufacturers have been successful in producing very tightly sealed light fixtures and electronics enclosures, this has still not prevented water incursion.

One type of electrical connector is known as a Deutsch Connector. Deutsch Connectors are environmentally sealed, waterproof electrical connectors designed for the transportation industry. The rugged thermoplastic housings operate in temperatures from −55° C. to 125° C. and include Silicone Rubber seals. Deutsch Connectors are available with a variable number of cavities depending on the application. An example of various Deutsch Connectors can be seen at https://www.deutschconnector.com/products/deutsch_connectors/.

In some configurations, Deutsch Connectors are provided as flange mounted connectors as can be seen at https://www.deutschconnector.com/products/deutsch_connectors/deutsch_flange_mount_connectors/. However, a major problem with flange mounted Deutsch Connectors is that they tend to break. In rugged environments, such as in the automotive industry, the plugging and unplugging of a Deutsch Connector can be difficult and requires a person to pull very hard to withdraw the male connector from the flange connector. Quite often, a person will wiggle the connector from side to side as they try to work the male connector loose from the flange body. This quite often leads to cracking of the flange body as they are formed as a single unitary piece typically made from a hard thermoplastic material. Additionally, for flange mounted connectors that are affixed to a surface, if the screws that hold the Deutsch Connector are over-tightened, this too can crack the flange connector.

It should be noted that the Deutsch Connector has been specifically selected to be formed with a single unitary structure formed of thermoplastic material. This is important because this construction limits the water intrusion points for the connector.

When a Deutsch Connector is damaged, the entire flange connector body needs to be removed and replaced. This typically involves removing the flange connector body from the surface it is connected to, pulling the wires through the opening, and then cutting the wires that have previously been soldered to the equipment leads (e.g., light fixture, electronics equipment, etc.). Non-soldered connections are typically not acceptable because of the ruggedness of the environment. For example, if the application is a light bar on a vehicle, the vibration of the vehicle when being driven will cause connections such as wire nuts to come loose causing the equipment to malfunction. Accordingly, replacing a broken Deutsch Connector is a time and labor-intensive process.

Additionally, the lamps used in some types of light fixtures can generate a significant amount of heat. When heat is generated by the lamp, it functions to heat the surrounding air inside the sealed light fixture, which in turn causes the air to expand. However, in light fixtures with very tight seals, the heated air is unable to easily escape the interior of the fixture and therefore the expanding air increases the air pressure inside the fixture. The pressurized air inside the light fixture then seeks equilibrium with the lower pressure air outside the fixture using any pathway available. One pathway is the air space inside the electrical cables (the space between the electrical conductor strands and space between the electrical conductors and the surrounding insulation). The result has been that air inside the fixture passes through the electrical cable and escapes the interior of the light fixture through the electrical fittings. However, while escaping air is not necessarily problematic, when the light fixture is turned off and the lamp inside cools down, the air inside the light fixture also cools down and contracts causing a negative pressure to develop inside the light fixture. The negative pressure functions to draw air and moisture into the light fixture through the same path that air escaped, namely, via the electrical cable and fittings. When using a Deutsch Connector, the pathway extends through the male connector and through the connector body itself. Over time, the repeated cycles of heating and cooling can cause a significant amount of moisture to be drawn into the light fixture. This moisture in turn, is subject to the heating and cooling when the light fixture turns on/off, which leads to condensation throughout the light fixture leading to accelerated corrosion, degradation, and eventual premature failure of the equipment.

SUMMARY OF THE INVENTION

What is desired then is an electrical connector that can be mounted to a surface that prevents water from passing through the electrical connector.

It is also desired to provide an electrical connector that can be mounted to a surface that resists breaking or cracking of the electrical connector.

It is further desired to provide an electrical connector that can be mounted to a surface that is provided as a two-piece structure where the portion of the connector that forms a cavity is detachable from the portion that is mounted to the surface.

It is still further desired to provide an electrical connector that can be mounted to a surface that is resistant to breakage and is adapted to receive a Deutsch Connector.

It is also desired to provide a watertight electrical connector that comprises a two-piece structure that reduces the time and difficulty of assembly.

It is also desired to provide an electrical connector that can be plugged directly into a socket provided in the surface of a light fixture where the electrical connector provides a watertight seal in and around the socket and through the interior of the electrical connector.

In one configuration an electrical connector is provided that is insertable into a socket, which may be provided in the housing of a light fixture. The electrical connector may be formed as a two-piece connector comprising a connector base and a connector body. In one configuration, the connector base is formed of an elastic material, while the connector body that is adapted to sit on top of the connector base is formed of a rigid metal material.

The connector base, in one configuration, is formed as a single integral piece and is formed having a body portion and a flange section. The body portion is adapted to be inserted into an opening formed in a surface (e.g., into a socket formed in a housing), which could be, for example, a housing of a light fixture or a mounting surface in a vehicle engine compartment. The body portion may in one configuration, be provided with a series of ridges or raised portions extending circumferentially around the body portion. When inserted into the opening/socket, the series of ridges or raised portions interact with an interior surface of the opening/socket forming a seal preventing the ingress of water through the opening. In one configuration, pins may extend from the body portion that are designed to interact with corresponding receptacles in the socket. The pins may be formed of a conductive material and are sealed around the end of the body portion preventing water from passing through or around the edges of the pins.

The connector base may further comprise a flange section that is wider than the body portion where the flange section interacts with the surface surrounding the opening (e.g., the surface of the housing) allowing the connector base to be securely attached to the housing. In one configuration, the flange section may be provided with two openings each opposite to each position outward from the body portion and adapted to receive a fastening member, such as a screw or the like that can secure the electrical connector in the socket preventing accidental removal of the electrical connector.

The connector body, in one configuration, is formed as a single integral piece and is formed having a flange portion and a raised section. The flange portion of the connector body is adapted to interact with the flange section of the connector base. The flange portion and the flange section interact with each other such that multiple seals are formed preventing water from passing from between the flange portion and the flange section. The flange portion of the connector body is further provided in one configuration with two openings corresponding to the two openings in the flange section allowing respective fastening members to pass through both the flange section and the flange portion. When the fastening members are tightened, the flange portion is compressed onto the flange section, which in turn is compressed to the surface.

The connector body further includes a raised section that forms a cavity. The cavity is adapted to receive a male connector that in one configuration comprises, a Deutsch Connector. The connector body may, in one configuration, be formed from aluminum. Additionally, the raised section is formed with a locking mechanism adapted to interact with an arm associated with a male connecter. The locking mechanism may comprise at least one opening in the raised section adapted to receive a protrusion on the arm. In practice, when the male connector is inserted into the cavity, the protrusion on the arm will contact an inner surface of the cavity causing the arm to deflect inward. When the male connector is fully advanced into the cavity, the protrusion axially aligns with the at least one opening allowing the arm to deflect outward such that the protrusion extends through the at least one opening. This in turn, locks the male connector into the cavity preventing accidental removal of the male connector from the electrical connector.

It should further be noted that the connector base and the connector body are keyed such that the connector body can only be attached to the connector base in one direction ensuring that the electrical connector forms a proper seal between the two pieces. Likewise, the socket is keyed so that the electrical connector can only be inserted into the socket in the correct orientation.

It will further be understood by those of skill in the art that any number of pins may be provided in the electrical connector (e.g., 2 pins, 3 pins, 4 pins, etc.), which would further include corresponding electrical conductors. The pins in the connector body will correspond to the pins extending from the connector base that are designed to engage with receptacles positioned in the socket. The pins may be provided that extend completely through the electrical connector. For example, the pins may extend from a distal end of the electrical connector (e.g., in the cavity of the Deutsch Connector), they may extend through the electrical connector, and extend from a proximal end of the electrical connector (e.g., designed to engage with a socket in the housing of a light fixture). This configuration simplifies the assembly process as there is no soldering to perform and there are no wires to feed through internal openings in the electrical connector. The pins are provided as a single unitary structure as indicated in the Figures and may include a seat having a shoulder designed to hold the pins in a predefined position within the electrical connector.

In one configuration the receptacles at the bottom end of the socket may be coupled to a circuit board such that when the electrical connector is inserted in the proper orientation into the socket, the pins extending from the connector base align with and engage with corresponding receptacles as the electrical connector is advanced into the socket. Once the electrical connector is fully advanced into the socket, the flange portion of the connector base will sit flat against a surface of the light fixture housing with the pins fully advanced into the receptacles. Additionally, a seal is formed on an undersurface of the flange between the connector base and the surface of the housing. Additional seals are provided by the series of ridges or raised portions extending circumferentially around the insertion portion that interact with the interior surface of the socket forming a series of seals.

In one configuration, the internal structure of the electrical connector is described. Electrical conductors extend through the base portion connecting at a distal end to the pins extending from an upper surface of the base section and at a proximal end to the pins extending from a bottom surface of the base section. The electrical conductors, which may comprise copper stranded conductors, may be coupled to the sets of pins via a solder connection inside the body portion. The electrical conductors may have the electrical insulation surrounding the electrical conductors removed from the ends of the electrical conductors and for a longitudinal distance, such as but not limited to, 1 inch. Solder may be applied to the ends of the electrical conductors to bond the electrical conductors to the pins. Additionally, the solder may be applied to the stranded conductors over the area where the insulation has been removed (i.e., the electrical insulation could be completely removed from the conductors inside the connector base). This effectively converts the stranded conductors to solid conductors for that section of the wire. The interior of the base portion can then be filled with an epoxy surrounding the ends of the pins, the solder joints, and the portion of the electrical conductors within the base portion. This configuration will effectively prevent any water from traveling along the electrical conductors as the solid portions of the electrical conductors will prevent water from passing through spaces between the stranded wires and the epoxy will prevent any water from traveling along a surface of the soldered electrical conductors.

For this application the following terms and definitions shall apply:

The terms “first” and “second” are used to distinguish one element, set, data, object or thing from another, and are not used to designate relative position or arrangement in time.

The terms “coupled”, “coupled to”, “coupled with”, “connected”, “connected to”, and “connected with” as used herein each mean a relationship between or among two or more devices, apparatus, files, programs, applications, media, components, networks, systems, subsystems, and/or means, constituting any one or more of (a) a connection, whether direct or through one or more other devices, apparatus, files, programs, applications, media, components, networks, systems, subsystems, or means, (b) a communications relationship, whether direct or through one or more other devices, apparatus, files, programs, applications, media, components, networks, systems, subsystems, or means, and/or (c) a functional relationship in which the operation of any one or more devices, apparatus, files, programs, applications, media, components, networks, systems, subsystems, or means depends, in whole or in part, on the operation of any one or more others thereof.

In one configuration watertight electrical connector is provided comprising, a connector base including: a body portion adapted to be inserted into an opening of a surface and forming a seal between an inner surface of the opening and an exterior surface of the body portion, and a flange section adapted to engage with the surface and having at least one hole adapted to receive a fastening member. The electrical connector further comprises, a connector body having: a flange portion adapted to interact with the flange section and form a seal between the connector body and the connector base, the flange portion having at least one opening adapted to receive the fastening member, and a raised section forming a cavity adapted to receive a male connector. The electrical connector further comprises at least two conductors entering a proximal end of the body portion and coupled to corresponding pins mounted on the flange section. The electrical connector also comprises, a spacer having at least two openings therein to allow the at least two conductors to past through, the spacer inserted into the body portion to fill an internal space of the body portion, the spacer forming a seal between an outer surface of the spacer and an inner surface of the body portion, and an epoxy applied to at least a portion of the internal space such that the epoxy creates a seal preventing water from traveling through the body portion. The electrical connector is provided such that the connector body is detachably connectable with the body portion.

In another configuration a watertight electrical connector is provided comprising, a connector base including: a body portion adapted to be received in an opening of a surface and adapted to form a first seal between an inner surface of the opening and an exterior surface of the body portion, and a flange section having a raised area and a shoulder each positioned on a distal surface of the flange section, the flange section adapted to engage with the surface. The electrical connector further comprises, a connector body having: a flange portion having a shoulder and an upstanding wall extending around a perimeter of the flange portion, and a raised section forming a cavity adapted to receive a male connector. The electrical conductor is provided such that the raised area on the distal surface of the flange section interacts with the shoulder of the flange portion such that a second seal is formed between the cavity and the distal surface. The electrical conductor is also provided such that the upstanding wall of the flange portion interacts with the shoulder of the flange section to form a third seal between the flange portion and the flange section. The electrical connector further comprises at least two conductors entering a proximal end of the body portion and coupled to corresponding pins mounted on the raised area. The electrical connector also comprises, a spacer having at least two openings therein to allow the at least two conductors to pass through, the spacer inserted into the body portion to fill an internal space of the body portion, the spacer forming a fourth seal between an outer surface of the spacer and an inner surface of the body portion, and an epoxy applied to at least a portion of the internal space such that the epoxy creates a fifth seal preventing water from traveling through the body portion. The electrical conductor is still further provided such that the connector body is detachably connectable with the body portion.

Other objects of the invention and its features and advantages will become more apparent from consideration of the following drawings and accompanying detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is an exploded view of an electrical connector utilizing a four-pin configuration according to one configuration of the invention.

FIG. **2** is a perspective view of the connector base according to FIG. **1**.

FIG. **3** is another perspective view of the connector base according to FIG. **1**.

FIG. **4** is a front view of the connector base mounted to a surface according to FIG. **1**.

FIG. **5** is a rear view of the connector base mounted to a surface according to FIG. **1**.

FIG. **6** is a rear view of the connector body according to FIG. **1**.

FIG. **7** is a perspective view of the connector base mounted to a surface and the connector body according to FIG. **1**.

FIG. **8** is a front view of the electrical connector mounted to a surface according to FIG. **1**.

FIG. **9** is a perspective view of the electrical connector mounted to a surface according to FIG. **8**.

FIG. **10** is an exploded view of an electrical connector utilizing a three-pin configuration according to one configuration of the invention.

FIG. **11** is a perspective view of the connector base according to FIG. **10**.

FIG. **12** is another perspective view of the connector base according to FIG. **10**.

FIG. **13** is a perspective view of the connector body according to FIG.

FIG. **14** is a perspective view of a male connector that can be inserted into the electrical connector according to FIG. **10**.

FIG. **15** is an exploded view of an electrical connector utilizing a two-pin configuration according to one configuration of the invention.

FIG. **16** is a front perspective view of the connector base mounted to a surface according to FIG. **15**.

FIG. **17** is a perspective view of the connector body according to FIG. **15**.

FIG. **18** is a front perspective view of the electrical connector mounted to a surface according to FIG. **15**.

FIG. **19** is a perspective view of a male connector that can be inserted into the electrical connector according to FIG. **15**.

FIG. **20** is a top perspective view of an electrical connector utilizing a two-pin configuration and having a pin plugin configuration according to one configuration of the invention.

FIG. **21** is a bottom perspective view of the electrical connector according to FIG. **20**.

FIG. **22** is a top view of the electrical connector according to FIG. **20**.

FIG. **23** is a top perspective exploded view of the electrical connector according to FIG. **20**.

FIG. **24** is a top perspective view of an electrical connector utilizing a three-pin configuration and having a pin plugin configuration according to one configuration of the invention.

FIG. **25** is a bottom perspective view of the electrical connector according to FIG. **24**.

FIG. **26** is a top view of the electrical connector according to FIG. **24**.

FIG. **27** is a top perspective exploded view of the electrical connector according to FIG. **24**.

FIG. **28** is a top perspective view of an electrical connector utilizing a four-pin configuration and having a pin plugin configuration according to one configuration of the invention.

FIG. **29** is a bottom perspective view of the electrical connector according to FIG. **28**.

FIG. **30** is a top view of the electrical connector according to FIG. **28**.

FIG. **31** is a top perspective exploded view of the electrical connector according to FIG. **28**.

FIG. **32** is a side view of an electrical connector according to FIG. **28** illustrating a side of the housing of a light fixture and a circuit board that includes receptacles corresponding to the pins extending from the insertion section.

FIG. **33** is a top perspective view of an electrical connector according to FIG. **32** illustrating the socket configuration and a circuit board that includes receptacles corresponding to the pins extending from the insertion section.

DETAILED DESCRIPTION OF THE INVENTION

Referring now to the drawings, wherein like reference numerals designate corresponding structure throughout the views.

FIG. **1** is an exploded view of the electrical connector **100** according to one configuration of the invention. The electrical connector **100** comprises a connector base **102** and a connector body **104**. The connector body **104** is detachable from the connector base **102** as shown in FIG. **1**.

The connector base **102** is made from an elastic material, such as, a thermal plastic rubber (TPR), neoprene, thermoplastics elastomers (TPE), silicon, flexible polyvinyl chloride (PVC), and the like.

The connector base **102** is designed to fit into an opening **10** of a surface **12** as shown in FIGS. **2** & **3**. A generic surface is shown in the drawings to illustrate how the electrical connector **100** can be affixed to a surface **12**.

The connector base **102** comprises a body portion **106** and a flange section **108**, which can variously be seen in FIGS. **1-3**. Body portion **106** has an exterior surface **110**, that includes a series of upstanding ridges **112**. The upstanding ridges **112** are formed of the elastic material and are designed to be slightly larger in diameter than the opening **10** such that they deform when the body portion **106** is inserted into the opening **10** forming a first seal for the electrical connector **100**. While five upstanding ridges **112** have been used in the current example, it is contemplated the fewer or more may be used depending on the application.

The connector base **102** also includes flange section **108** that is formed having a perimeter larger than the body portion **106**. The flange section **108** includes a contact surface **114** that is designed to lay flat against surface **12** (FIGS. **3** & **4**). Additionally, flange section **108** is formed having two elongated sections opposite each other having holes **116**, **118** formed therein for receiving fastening members **120**, **122** respectively as shown in FIG. **1**.

The flange section **108** also includes a raised area **124** on an upper surface **126** of the flange section **108**. The raised area **124** forms a step at a perimeter of the raised area **124**. Also shown in FIGS. **2-4** are pins **132**, which extend from holes **134** in the raised area **124**.

The flange section **108** further includes a shoulder **128** formed around a perimeter **130** of the flange section **108**. Shoulder **128** is formed as a step as can be seen in FIGS. **1-3**.

In one configuration, the entire connector base **102** is formed as a single integral piece of elastic material.

The connector base **102** and the connector body **104** are keyed such that the connector body **104** can only be attached to the connector base **102** in one direction ensuring that the electrical connector **100** forms a proper seal between the two pieces. The connector body **104** includes a key **170** that is designed to interact with a corresponding key **172** formed on connector base **102**.

Electrical conductors **136**, which may comprise copper (Cu) stranded conductors, are connected to pins **132**. In one configuration, the connection between the electrical conductors **136** and pins **132** is formed by soldering. Also shown is an area where electrical insulation **138** is removed from the electrical conductors **136** to form a section of bare wire **140**. The bare wire may have a solder applied to it so that the section of bare wire effectively becomes a “solid” wire where the solder is applied. The pins **132**, extend through openings **134** to extend substantially perpendicular to the raised area **124**, and the area of “solid” wire and the location where the electrical conductor **136** is soldered to the end of the pin **132** is positioned within the interior of the body portion **106**.

Also shown in FIG. **1** is spacer **154** that includes openings **156** designed to receive electrical conductors **136**. The exterior surface of spacer **154** includes a series of upstanding ridges **158** that are designed to interact with an inner surface of a cavity (not shown) in body portion **106**. Spacer **154** comprises the elastic material as described in connection with connector base **102**. The electrical conductors are threaded through the openings **156** with the pins **132** extending therefrom. Spacer **154** is then inserted into the cavity (not shown) in body portion **106** such that the pins **132** are threaded through openings **134** of raised area **124**.

In one configuration, an epoxy **142** (FIG. **5**) may be applied to the interior space of the body portion **106** to completely fill up any space left around the ends of the pins, the soldered joint, the "solid" wire, and the transition to the stranded wire. Additionally, epoxy **142** may be applied to completely cover the end **160** of spacer **154**.

The epoxy may be selected as a thermal conductive epoxy with high electrical resistance characteristics. In this way, water is completely prevented from traveling through the body portion **106** whether through or around the electrical conductors **136**, or around the pins **132**.

Referring now to FIGS. **1** & **6-9**, connector body **104** is variously illustrated. As indicated in FIGS. **1**, **6** & **7**, the connector body **104** is detachable from the connector base **102**. In one configuration, the connector body **104** is formed of aluminum, however, it is contemplated that other metal materials may effectively be utilized.

The connector body **104** comprises a flange portion **144** and a raised section **146**. The flange portion **144** has a perimeter that is essentially equal to the perimeter of the flange section **108** and is designed to sit on top of the flange section **108**.

Referring now to FIG. **6**, an upstanding wall **148** extends around the perimeter of flange portion **144**. When the flange portion **144** is fit over top of the flange section **108**, the upstanding wall **148** interacts with the shoulder **128** to form a second seal around an outer edge of connector base **102** and connector body **104** for the electrical connector **100**.

Also illustrated in FIG. **6** is shoulder **150** extends around a proximal end of a cavity **152** formed by raised section **146**. When the flange portion **144** is fit over top of the flange section **108**, an outer edge of raised area **124** interacts with shoulder **150** to form a third seal between an outer edge of the cavity **152** of raised section **146** and the and the outer edge of raised area **124** of flange section **108** for the electrical connector **100**.

A locking mechanism (FIGS. **1**, **6** & **8**) is variously illustrated comprising a channel **164** formed in the raised section **146**, which is provided with two openings **166**, **168**, provided therein. These two openings **166**, **168** may be provided as elongated channels and are designed to interact with two protrusions **22**, **24** formed on an arm **26** provided on a male connector **20**.

The male connector **20** may be provided with an exterior surface **28** that may include at least one or more O-ring(s) **30** that can interact with an interior surface of cavity **152** formed by raised section **146**. Arm **26** is designed to engage with channel **164** formed in the raised section **146**. When the male connector **20** is fully advanced into electrical connector **100**, the protrusions **22**, **24** will interlock with openings **166**, **168** for a lock such that the male connector **20** cannot be withdrawn without pressing downward on arm **26** to disengage the protrusions **22**, **24** from openings **166**, **168**.

An end face **38** of male connector **20** is provided with four receptacles **32** that are designed to receive pins **132**. In this configuration, there are four pins **132** and four corresponding receptacles **32**.

Referring to FIG. **10**, an exploded view of the electrical connector **200** according to one configuration of the invention is presented. The electrical connector **200** comprises a connector base **202** and a connector body **204**. The connector body **204** is detachable from the connector base **202** as shown in FIG. **10**.

The connector base **202** may comprise an elastic material as described in connection with the 4-pin configuration previously described.

The connector base **202** is designed to fit into an opening **10** of a surface **12** as shown in FIGS. **12**, **16** & **18**. A generic surface is shown in the drawings to illustrate how the electrical connector **200** can be affixed to a surface **12**.

Connector base **202** comprises a body portion **206** and a flange section **208**, which can variously be seen in FIGS. **11** & **12**. Body portion **206** has an exterior surface **210**, that includes a series of upstanding ridges **212**. The upstanding ridges **212** are formed of the elastic material and are designed to be slightly larger in diameter than the opening **10** such that they deform when the body portion **206** is inserted into the opening **10** forming a first seal for the electrical connector **200**. While five upstanding ridges **212** have been used in the current example, it is contemplated the fewer or more may be used depending on the application.

The connector base **202** also includes flange section **208** that is formed having a perimeter larger than the body portion **206**. Flange section **208** includes contact surface **214** that is designed to lay flat against surface **12** (FIG. **14**). Additionally, flange section **208** is formed having two elongated sections opposite each other having holes **216**, **218** formed therein for receiving fastening members (not shown).

Flange section **208** also includes a raised area **224** on an upper surface **226** of the flange section **208**. The raised area **224** forms a step at a perimeter of the raised area **224**. Also shown in FIGS. **10** & **11** are pins **232**, which extend from holes **234** in the raised area **224**.

Flange section **208** further includes shoulder **228** formed around a perimeter **230** of the flange section **208**. The shoulder **228** is formed as a step as can be seen in FIGS. **10-12**.

In one configuration, the entire connector base **202** is formed as a single integral piece of elastic material.

The connector base **202** and the connector body **204** are keyed such that the connector body **204** can only be attached to the connector base **202** in one direction ensuring that the electrical connector **200** forms a proper seal between the two pieces. The keys are similar to those described in connection with the four-pin configuration.

Electrical conductors **236** may comprise copper and may further include a soldered section as previously described. In one configuration, the connection between the electrical conductors **236** and pins **232** is formed by soldering. Also shown is an area where electrical insulation **238** is removed from the electrical conductors **236** to form a section of bare wire **240**. The bare wire may have a solder applied to it so that the section of bare wire effectively becomes a "solid" wire where the solder is applied. The pins **232**, extend through openings **234** to extend substantially perpendicular to the raised area **224**, and the area of "solid" wire and the location where the electrical conductor **236** is soldered to the end of the pin **232** is positioned within the interior of the body portion **206**.

Also shown in FIG. **10** is spacer **254** that includes openings **256** designed to receive electrical conductors **236**. The exterior surface of spacer **254** includes a series of upstanding ridges **258** that are designed to interact with an inner surface of a cavity (not shown) in body portion **206**. Spacer **254** comprises the elastic material as described in connection with connector base **202**. The electrical conductors are threaded through the openings **256** with the pins **232** extending therefrom. Spacer **254** is then inserted into the cavity (not shown) in body portion **206** such that pins **232** are threaded through openings **234** of raised area **224**.

In one configuration, an epoxy which may comprise a material as previously described, may be applied to the interior space of the body portion **206** to completely fill up any space left around the ends of the pins, the soldered joint, the "solid" wire, and the transition to the stranded wire. Additionally, the epoxy may be applied to completely cover the end of spacer **254**.

Referring now to FIGS. **10**, **13**, **18** & **19**, connector body **204** is variously illustrated. The connector body **204** is detachable from connector base **202**. In one configuration, connector body **204** is formed of aluminum, however, it is contemplated that other metal materials may effectively be utilized.

Connector body **204** comprises a flange portion **244** and a raised section **246**. The flange portion **244** has a perimeter that is essentially equal to the perimeter of flange section **208** and is designed to sit on top of flange section **208**.

Referring now to FIG. **13**, an upstanding wall **248** extends around the perimeter of flange portion **244**. When the flange portion **244** is fit over top of the flange section **208**, the upstanding wall **248** interacts with the shoulder **228** to form a second seal around an outer edge of connector base **202** and connector body **204** for the electrical connector **200**.

Also illustrated in FIG. **6** is shoulder **250** extends around a proximal end of cavity **252** formed by raised section **246**. When the flange portion **244** is fit over top of the flange section **208**, an outer edge of raised area **224** interacts with shoulder **250** to form a third seal between an outer edge of the cavity **252** of raised section **246** and the and the outer edge of raised area **224** of flange section **208** for the electrical connector **200**.

A locking mechanism (FIGS. **13**, **14**) is variously illustrated comprising a channel **264** formed in the raised section **246**, which is provided with an opening **266**, provided therein. These opening **266** may be provided as elongated channels and are designed to interact with protrusion **52** formed on an arm **56** provided on a male connector **50**.

The male connector **50** may be provided with an exterior surface **58** that may include at least one or more O-ring(s) **60** that can interact with an interior surface of cavity **252** formed by raised section **246**. Arm **56** is designed to engage with channel **264** formed in the raised section **246**. When the male connector **50** is fully advanced into electrical connector **200**, the protrusion **52** will interlock with opening **266** to lock such that the male connector **50** cannot be withdrawn without pressing downward on arm **56** to disengage the protrusion **52** from opening **266**.

An end face (not shown) of male connector **50** is provided with three receptacles (not shown) that are designed to receive pins **232**.

FIG. **15** is an exploded view of the electrical connector **300** according to one configuration of the invention. The electrical connector **300** comprises a connector base **302** and a connector body **304**. The connector body **304** is detachable from the connector base **302** as shown in FIG. **15**.

The connector base **302** may comprise an elastic material as described in connection with the 4-pin configuration previously described.

The connector base **302** is designed to fit into an opening of a surface as previously described in connection with the four-pin and three-pin versions.

The connector base **302** comprises a body portion **306** and a flange section **308**, which can variously be seen in FIGS. **1-3**. The body portion **306** has an exterior surface **310**, that includes a series of upstanding ridges **312**. The upstanding ridges **312** are formed of the elastic material and are designed to be slightly larger in diameter than the opening in which they are inserted such that they deform when the body portion **306** is inserted into the opening forming a first seal for the electrical connector **300**.

The connector base **302** also includes flange section **308** that is formed having a perimeter larger than the body portion **306**. The flange section **308** includes a contact surface **314** that is designed to lay flat against surface **12** (FIGS. **16**, **18** & **19**). Additionally, the flange section **308** is formed having two elongated sections opposite each other having holes **316**, **318** formed therein for receiving fastening members (not shown) respectively.

The flange section **308** also includes a raised area **324** on an upper surface **326** of the flange section **308**. The raised area **324** forms a step at a perimeter of the raised area **324**. Also shown in FIGS. **15**, **16** & **18** are pins **332**, which extend from holes **334** in the raised area **324**.

The flange section **308** further includes a shoulder **328** formed around a perimeter **330** of the flange section **308**. The shoulder **38** is formed as a step as can be seen in FIGS. **15** & **16**.

In one configuration, the entire connector base **302** is formed as a single integral piece of elastic material.

The connector base **302** and the connector body **304** are keyed such that the connector body **304** can only be attached to the connector base **302** in one direction ensuring that the electrical connector **300** forms a proper seal between the two pieces. The connector body **304** includes a key (not shown) that is designed to interact with a corresponding key **372** formed on connector base **302**.

Electrical conductors **336**, which may comprise copper (Cu) stranded conductors, are connected to pins **332**. In one configuration, the connection between the electrical conductors **336** and pins **332** is formed by soldering. Also shown is an area where electrical insulation **338** is removed from the electrical conductors **336** to form a section of bare wire **340**. The bare wire may have a solder applied to it so that the section of bare wire effectively becomes a "solid" wire where the solder is applied. The pins **332**, extend through openings **334** to extend substantially perpendicular to the raised area **324**, and the area of "solid" wire and the location where the electrical conductor **336** is soldered to the end of the pin **332** is positioned within the interior of the body portion **306**.

Also shown in FIG. **15** is spacer **354** that includes openings **356** designed to receive electrical conductors **336**. The exterior surface of spacer **354** includes a series of upstanding ridges **358** that are designed to interact with an inner surface of a cavity (not shown) in body portion **306**. Spacer **354** comprises the elastic material as described in connection with connector base **302**. The electrical conductors are threaded through the openings **356** with the pins **332** extending therefrom. Spacer **354** is then inserted into the cavity (not shown) in body portion **306** such that the pins **332** are threaded through openings **334** of raised area **324**.

In one configuration, an epoxy which may comprise a material as previously described, may be applied to the interior space of the body portion **306** to completely fill up any space left around the ends of the pins, the soldered joint, the "solid" wire, and the transition to the stranded wire. Additionally, the epoxy may be applied to completely cover the end of spacer **354**.

Referring now to FIGS. **15** & **17-19**, the connector body **304** is variously illustrated. The connector body **304** is detachable from the connector base **302**. In one configuration, the connector body **304** is formed of aluminum, however, it is contemplated that other metal materials may effectively be utilized.

The connector body **304** comprises a flange portion **344** and a raised section **346**. The flange portion **344** has a perimeter that is essentially equal to the perimeter of flange section **308** and is designed to sit on top of the flange section **308**.

Referring now to FIG. **17**, an upstanding wall **348** extends around the perimeter of flange portion **344**. When the flange portion **344** is fit over top of the flange section **308**, the upstanding wall **348** interacts with the shoulder **328** to form a second seal around an outer edge of connector base **302** and connector body **304** for the electrical connector **300**.

Also illustrated in FIG. **17** is shoulder **350** extends around a proximal end of a cavity **352** formed by raised section **346**. When the flange portion **344** is fit over top of the flange section **308**, an outer edge of raised area **324** interacts with shoulder **350** to form a third seal between an outer edge of the cavity **352** of raised section **346** and the and the outer edge of raised area **324** of flange section **308** for the electrical connector **300**.

A locking mechanism (FIGS. **15**, **17-19**) is variously illustrated comprising a channel **364** formed in the raised section **346**, which is provided with two openings **366**, **368**, provided therein. These two openings **366**, **368** may be provided as elongated channels and are designed to interact with two protrusions **72**, **74** formed on an arm **76** provided on a male connector **70**.

The male connector **70** may be provided with an exterior surface **78** that may include at least one or more O-ring(s) **80** that can interact with an interior surface of cavity **352** formed by raised section **346**. Arm **76** is designed to engage with channel **364** formed in the raised section **346**. When the male connector **70** is fully advanced into electrical connector **300**, the protrusions **72**, **74** will interlock with openings **366**, **368** forming a lock such that the male connector **70** cannot be withdrawn without pressing downward on arm **76** to disengage the protrusions **72**, **74** from openings **366**, **368**.

An end face (not shown) of male connector **70** is provided with two receptacles (not shown) that are designed to receive pins **332**. In this configuration, there are two pins **332** and four corresponding receptacles (not shown).

Referring now to FIGS. **20-23**, an alternate configuration of the device described in connection with FIGS. **15-18** is provided. The electrical connector **300** is structurally similar to the two-pin configuration previously described, however, rather than having electrical conductors **336**, **338** extending from the body portion **306**, the configuration in FIGS. **20-23** illustrates pins **384** extending from an end face of the body portion **306**.

As can be seen with reference to FIG. **23**, pins **332** are insertable through holes **334** in the raised area **324**. The pins **332** each comprise a seat **380** having a flange **382** that is designed to sit against an underside of hole **334** to securely hold the pin **332** in the hole **334** when the spacer **354** is fully inserted.

The configuration in FIG. **23** differs from the configuration in FIG. **15** in that, rather than providing electrical conductors extending through the spacer **354**, pins **384** are provided. Once the pin structures are inserted and the spacer **354** is fully inserted into the connector base **302**, an epoxy can be applied to fill in any air space in the interior of body portion **306**.

Referring now to FIGS. **24-27**, an alternate configuration of the device described in connection with FIGS. **10-14** is provided. The electrical connector **200** is structurally similar to the three-pin configuration previously described, however, rather than having electrical conductors **236**, **238** extending from the body portion **206**, the configuration in FIGS. **24-27** illustrates pins **284** extending from an end face of the body portion **206**.

As can be seen with reference to FIG. **27**, pins **232** are insertable through holes **234** in the raised area **224**. The pins **232** each comprise a seat **280** having a flange **282** that is designed to sit against an underside of hole **234** to securely hold the pin **232** in the hole **234** when the spacer **254** is fully inserted.

The configuration in FIG. **27** differs from the configuration in FIG. **10** in that, rather than providing electrical conductors extending through spacer **254**, pins **284** are provided. Once the pin structures are inserted and spacer **254** is fully inserted into connector base **202**, an epoxy can be applied to fill in any air space in the interior of body portion **206**.

Referring now to FIGS. **28-33**, an alternate configuration of the device described in connection with FIGS. **1-9** is provided. The electrical connector **100** is structurally similar to the four-pin configuration previously described, however, rather than having electrical conductors **136**, **138** extending from the body portion **106**, the configuration in FIGS. **28-33** illustrates pins **184** extending from an end face of the body portion **106**.

As can be seen with reference to FIG. **31**, the pins **132** are insertable through holes **134** in the raised area **124**. The pins **132** each comprise a seat **180** having a flange **182** that is designed to sit against an underside of hole **134** to securely hold the pin **132** in the hole **134** when the spacer **154** is fully inserted.

The configuration in FIG. **31** differs from the configuration in FIG. **1** in that, rather than providing electrical conductors extending through spacer **154**, pins **184** are provided. Once the pin structures are inserted and spacer **154** is fully inserted into connector base **102**, an epoxy can be applied to fill in any air space in the interior of body portion **106**.

Referring now to FIGS. **32** and **33**, a socket **90** provided in a housing **92** (only a portion of the housing **92** is shown) is illustrated along with the body portion **106** that is insertable into the socket **90**. Pins **184** extend from body portion **106** and are designed to interact with openings **94** (in this example there are four openings) comprising a receptacle that in this configuration is positioned on printed circuit board (PCB) **96**. While the openings **92** are shown in this configuration on PCB **96**, it is contemplated that the receptacle could comprise a separate plug configuration provided in socket **90** that is electrically connected to PCB **94**. Additionally, the number of openings will correspond to the connector that is used (i.e., a two-pin, three-pin or four-pin configuration). While 2, 3 and 4 pin configurations are described, it is contemplated that any number of pins may effectively be used without deviating from the invention.

Body portion **106** includes an exterior surface **110**, that includes a series of upstanding ridges **112**. The upstanding ridges **112** are formed of elastic material and are designed to be slightly larger in diameter than an inner surface **98** of socket **90**. In this way, the upstanding ridges **112** deform when the body portion **106** is inserted into the socket **90** forming five separate seals for the electrical connector **100**. It should be noted however, that the depth of the socket may be variable and that not all five seals may be needed. For example, the socket could be much shallower than depicted in FIGS. **32** and **33** and only one or two seals may be formed between the inner surface **98** and the upstanding ridge(s).

Also provided on electrical connector **100** is a contact surface **190** (FIG. **32**) that is designed to sit flat against an exterior surface **98** of the light fixture housing. The contact surface may be formed of an elastic material such that when the electrical connector is fully inserted into socket **90** a seal is formed between the contact surface **190** and the exterior surface **97** of the light fixture housing. In some configurations, screws may be used to firmly hold the electrical connector **100** in a fully inserted position with socket **90**. In other configurations, the screws may not be needed as the seals between the upstanding ridges **112** and the inner surface **98** of the socket as well as the frictional fit of the pins **180** with the openings **92** will create a secure connection. However, in certain applications where vibration may be intense, securing the electrical connector **100** by means of locking mechanism (e.g., screws or the like) is preferable.

Although the invention has been described with reference to a particular arrangement of parts, features and the like, these are not intended to exhaust all possible arrangements or features, and indeed many other modifications and variations will be ascertainable to those of skill in the art.

What is claimed is:

1. A watertight electrical connector comprising:
- a connector base including:
  - a body portion adapted to be inserted into an opening in a surface and forming a seal between an inner surface of the opening and an exterior surface of the body portion;
  - a flange section adapted to interact with the surface;
- a connector body having:
  - a flange portion adapted to interact with the flange section and form a seal between said connector body and said connector base;
  - a raised section forming a cavity adapted to receive a male connector;
- at least two pins mounted on the flange section;
- a spacer holding at least two conductors connected to said at least two pins respectively within an internal space of said body portion;
- an epoxy applied to the internal space such that said epoxy creates a seal preventing water from traveling through said body portion;
- wherein said connector body is detachablely connectable with said body portion; and
- wherein said connector base is detachably connectable with said opening.

2. The watertight electrical connector of claim **1**, wherein said at least two pins and said at least two conductors are each formed as a single extended pin structure where each extended pin structure extends through said body portion.

3. The watertight electrical connector of claim **2**, wherein each of said at least two pins comprises a seat having a shoulder.

4. The watertight electrical connector of claim **3**, wherein
- said flange section comprises a raised area on a distal surface of said flange section, the raised area including at least two openings extending through said flange section for said at least two pins respectively;
- wherein said seat is adapted to engage with an inner surface of the at least two openings extending through said flange section; and
- wherein said shoulder is adapted to engage with a surface of said flange section surrounding the opening to preventing said pin from advancing into said opening beyond the shoulder.

5. The watertight electrical connector of claim **4**, wherein
- the pins extend through the at least two opening extending through said flange section and extend beyond a distal surface of the raised area of the flange section; and
- wherein the pins extending beyond the distal surface of the raised area are adapted to engage with openings in a distal end of a male connector.

6. The watertight electrical connector of claim **4**, wherein when said spacer is advanced into the interior of said body portion, the pins extend through corresponding holes in said spacer and a front edge of said spacer interacts with said shoulder to sandwich the shoulder between the surface of said flange section surrounding the opening and the front edge of said spacer preventing any longitudinal movement of the pins when the spacer is fully inserted into said body portion.

7. The watertight electrical connector of claim **6**, wherein multiple upstanding ridges provided on the exterior surface of said spacer interact with the inner surface of said body portion and the epoxy functions as a seal preventing water from passing through said body portion.

8. The watertight electrical connector of claim **4**, wherein said flange portion comprises a shoulder adapted to interact with the raised area on the distal surface such that a seal is formed between the cavity and the distal surface of said flange section.

9. The watertight electrical connector of claim **8**, wherein the shoulder of said flange portion is formed as a step.

10. The watertight electrical connector of claim **2**, wherein each extended pin structure extends beyond a proximal end of said body portion.

11. The watertight electrical connector of claim **10**, wherein said opening in the surface comprises a receptacle with at least two openings corresponding to said at least two pins extending from said body portion, wherein when said body portion is fully inserted into the opening, said at least two pins extending from said body portion are fully inserted into said the at least two openings forming an electrical connection between each pin and each opening the pin is inserted into respectively.

12. The watertight electrical connector of claim **11**, wherein multiple seals are formed between the inner surface of the opening and the exterior surface of the body portion by multiple upstanding ridges provided on the exterior surface of the body portion interacting with the inner surface of the opening.

13. The watertight electrical connector of claim **11**, wherein each opening of the receptacle is electrically connected to a printed circuit board.

14. The watertight electrical connector of claim **13**, wherein the opening in the surface comprises a socket in a housing of a light fixture.

15. The watertight electrical connector of claim **1**, wherein said flange section has at least one hole adapted to receive a fastening member and said flange portion has at least one opening adapted to receive the fastening member.

16. The watertight electrical connector of claim **1**, wherein said raised section of the connector body includes a locking mechanism adapted to interact with an arm associated with a male connecter.

17. The watertight electrical connector of claim **16**, wherein said locking mechanism includes at least one opening in the raised section adapted to receive a protrusion on the arm.

18. The watertight electrical connector of claim **16**, wherein when the male connector comprises a Deutsch Connector.

19. A watertight electrical connector comprising:

a connector base including:

a body portion adapted to extend into an opening of a surface and forming a seal between an inner surface of the opening and an exterior surface of the body portion;

a flange section adapted to engage with the surface and having at least one hole adapted to receive a fastening member;

a connector body having:

a flange portion adapted to interact with the flange section, said flange portion having at least one opening adapted to receive the fastening member;

a raised section forming a cavity adapted to receive a connector;

at least two pins extending at least partially through the body portion and into the cavity and adapted to be coupled to corresponding conductors;

the at least two pins held a fixed distance from each other such that the at least two pins are rigidly maintained in the cavity for engagement with a connector adapted to be inserted into the cavity;

wherein said connector body is detachablely connectable with said body portion.

20. The watertight electrical connector of claim 19, wherein the connector base comprises a rigid material.

21. The watertight electrical connector of claim 19, wherein

said flange portion comprises an upstanding wall extending around a perimeter of said flange portion; and

said flange section is adapted to interact with the upstanding wall such that a seal is formed between said flange portion and said flange section.

22. The watertight electrical connector of claim 19, wherein said flange section and said flange portion each have at least two openings adapted to receive respective fastening members.

23. The watertight electrical connector of claim 19, further comprising a raised section of the connector body having a locking mechanism adapted to interact with an arm associated with a connecter adapted to be inserted into the cavity.

24. The watertight electrical connector of claim 23, wherein the locking mechanism includes at least one opening in the raised section adapted to receive a protrusion on the arm.

25. The watertight electrical connector of claim 24, wherein when the connector is inserted into the cavity, the arm is deformed inward relative to the connector, and when the connector is fully advanced into the cavity the arm deflects outward relative to the connector such that the protrusion engages with the at least one opening in the raised section to lock the connector with the cavity.

26. The watertight electrical connector of claim 23, wherein when the connector comprises a Deutsch Connector.

27. The watertight electrical connector of claim 19, wherein the seal between the inner surface of the opening and the exterior surface of the body portion is formed by at least one upstanding ridge provided on the exterior surface of the body portion interacting with the inner surface of the opening.

28. The watertight electrical connector of claim 19, wherein said connector body is formed of an elastic material.

29. The watertight electrical connector of claim 28, wherein the elastic material is selected from the group consisting of: thermal plastic rubber (TPR), neoprene, thermoplastics elastomer (TPE), silicon, and flexible polyvinyl chloride (PVC).

30. The watertight electrical connector of claim 19, wherein said connector base is formed of an elastic material.

31. The watertight electrical connector of claim 30, wherein the elastic material is selected from the group consisting of: thermal plastic rubber (TPR), neoprene, thermoplastics elastomer (TPE), silicon, and flexible polyvinyl chloride (PVC).

32. The watertight electrical connector of claim 19, wherein the at least two pins comprise three or four pins, each pin having a corresponding conductor affixed thereto.

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